

The High-Frequency Factor Zoo

(Working Paper)

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September 23, 2022

Abstract

I construct a novel dataset of 224 high-frequency factor portfolios in order to study the cross-section of expected returns in a continuous-time setting. I estimate the continuous and semijump risk premia for each of these factors and find that jump and semijump risk are often priced and command a larger risk premia than continuous risk. Furthermore, there only a few clusters of factors, corresponding to less than a third of the zoo, with significant continuous and semijump risk premia. Additionally, I decompose cross-sectional variation in expected returns into variation from exposure to the continuous and jump factor risk. I find that the majority of cross-sectional variation comes from jump risk and that most stocks draw significant jump risk premia.

Keywords: Factors; asset pricing; high frequency data; jump risk premia.

JEL Codes: C55, C58, G11, G12

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1 Introduction

Why do some assets offer higher returns than others? The asset pricing literature has provided a general answer to this question: differences in exposure to systematic risk. However, the question of what variables or so-called “factors” best represent systematic risk has remained difficult to conclude, with the literature proposing over 300 factors thus far (Harvey & Liu, 2019; Harvey, Liu, & Zhu, 2016). Furthermore, past work using continuous-time models has revealed that factor risk is far more granular than has been assumed. For instance, much of the risk premia associated with the market is driven by its “jump” returns (i.e., discontinuous changes in a semimartingale process) and, in particular, its negative jumps rather than its continuous component (Bollerslev, Li, & Todorov, 2016). Recent work using high-frequency Fama-French (2015) factors has found similarly sharp heterogeneity in the continuous and jump risk premia of the aforementioned non-market factors (Aït-Sahalia, Jacod, & Xiu, 2021).¹ However, further work in this direction involving the remaining factors in the “zoo” remains undone.

Hence, in this paper, I pursue exactly this and explore the continuous and jump risk premia to a vast set of factors set forth by the literature. To this end, I assemble a novel dataset consisting of 25 years of high-frequency characteristic-sorted portfolio returns for 224 previously proposed factors. These portfolios are constructed in the usual way, sorting on the characteristics of thousands of stocks, but my use of high-frequency stock return data enables me to obtain a similarly higher resolution for the resulting portfolio returns as well. I then rely critically on these high-frequency returns to justify the use of “infill asymptotic” arguments that enable the identification and separation of factor jumps from their diffusive component. Moreover, the same arguments allow me to use the separated returns to estimate continuous and jump betas for each factor. I use the betas in cross-sectional regressions applied over a large time horizon, *à la* Fama-MacBeth (1973), to identify their corresponding risk premia, a procedure formally justified in a double asymptotic setting by Aït-Sahalia, Jacod, and Xiu (2021). The resulting continuous and jump risk premia estimates reveal the pricing implications of essentially novel risk factors based on the continuous and jump components of the many proposed factors.

My results broadly suggest that both continuous and semijump risk is priced but only for a relatively small subset of factors. Starting with the workhorse factor models based on the Fama-French 6 (FF5 + Momentum), I find that the only the jump component of market and the continuous component of the operating profitability portfolio (Robust-Minus-Weak) command statistically significant risk premia. Interestingly, nearly all of the excess return on the market, 6.3%, comes from jump risk premia while about 3.4% of the return comes from negative jump risk premia. Similarly, the continuous component of Robust-Minus-Weak draws a 4% risk premia, which is substantial relative to its in-sample average return. On the other hand, the risk premia associated with the remaining factors are statistically insignificant.

Next, with respect to the factor zoo, I similarly find that only a relatively small portion the many risk factors are associated with statistically significant risk premia – 42 out of

¹ For additional work on high-frequency multifactor models, see also Aït-Sahalia, Kalnina, and Xiu (2020), Chabi-Yo, Huggenberger, and Weigert (2021), and Pelger (2020).

654 estimates in total – after adjusting for multiple hypothesis testing. The signs of the estimates are as expected, with investors paying a premium for hedging against factor crashes as represented by negative jumps in the portfolios; likewise, positive jumps tend to draw a negative risk premia while the continuous risk premia are generally positive. The magnitudes of all three premia (continuous and the two semijump) are on par with the magnitudes of the portfolio returns themselves, suggesting that all three are relatively economically significant. However, one potential issue with an analysis of the entire zoo is that many factors tend to be non-trivially positively and negatively correlated with one another. This fact complicates multiple testing adjustments and the interpretation of the corresponding statistical results.

Thus, to tackle this issue, I also perform an analysis of “cluster portfolios” constructed using the first principal component of factor clusters based on Jensen, Kelly, and Pedersen (2021). These thirteen representative portfolios readily capture the common variation within each cluster while retaining the economic interpretability of the underlying factors. Interestingly, I find that only four risk factors based on these portfolios draw both robust and statistically significant risk premia: negative jumps in the Accruals cluster portfolio (1.64% p.a.), positive jumps in Skewness (-2.22 %) and Investment (-1.78) cluster portfolios, and continuous returns in Profitability cluster portfolio (3.01). These four clusters are associated with 5, 6, 23, and 15 factors from the zoo, implying that a non-trivial fraction of the many proposed factors are priced. Furthermore, consistent with the individual analysis of the zoo, the risk premia estimates for each of the three premia components are economically significant relative to the average return on the portfolios themselves, 1.12%. Evidence of non-trivial pricing implications for non-market jumps is also broadly consistent with past work by Jacod, Todorov, and Lin (2022) and Lin and Todorov (2019).

Overall, these results suggest that there exists clear heterogeneity in the continuous and (semi)jump risk premia of non-market factors. However, non-zero risk premia is a necessary but not sufficient condition for cross-sectional pricing ability. Hence, it remains unclear which is more important for explaining expected returns: continuous risk premia or jump risk premia. To address this, I introduce a novel variance decomposition to separately identify the proportion of cross-sectional variation explained by the continuous and jump components of the risk premia drawn by a given set of test assets. My decomposition follows naturally from the fact that covariation with the SDF, which is often proxied by a reduced-form factor covariances, can be clearly decomposed into continuous and jump (quadratic) covariations which themselves define the continuous and jump risk premia earned by an asset. My estimation methodology also follows readily from a modification of the Continuous-Time Fama-MacBeth regression proposed by Aït-Sahalia, Jacod, and Xiu (2021) that allows me to consistently estimate these two risk premia separately for any given test asset.

Implementing this decomposition on a set of large/mid-cap stocks and my high-frequency portfolios, I find clear evidence that the majority of cross-sectional variation in expected returns is explained by differences in exposure to systematic jump risk. That is, differences in covariation with the discontinuous component of the SDF explain why some assets have higher return than others. This result holds on the stocks and portfo-

lios separately, and it holds across a variety of factor models used to span the SDF. In contrast, I find that the continuous component of the SDF explains little variation in expected returns. Moreover, for most assets, the average risk premia earned from exposure to continuous risk is statistically insignificant. These results highlight both the importance of jump risk premia as well as the value in distinguishing jump and continuous returns when performing an analysis of systematic risk.

1.1 Literature Review

My results broadly connect three strands of literature in asset pricing. First is the asset pricing literature, which has produced an overwhelming number of additional factors to better model systematic risk.² There are now more than 300 factors (Harvey, Liu, & Zhu, 2016), raising concerns about multiple testing and p-hacking (Harvey, 2017). In response, a recent swath of papers has addressed this “factor zoo” (Cochrane, 2008) through a variety of approaches. One set of papers has replicated past factors both in-sample and out-of-sample; see, *e.g.*, Chen and Zimmermann (2020), Hou, Xue, and Zhang (2020), Jensen, Kelly, and Pedersen (2021), Pontiff and Woodgate, 2008. Some of these papers have also put an explicit focus on trading costs; see, *e.g.*, Chen and Velikov, 2021, Detzel, Novy-Marx, and Velikov, 2021, Frazzini, Israel, and Moskowitz, 2012, and Patton and Weller, 2020. On the other hand, another set of papers have introduced new econometric techniques to improve factor selection and reduce the dimensionality of the zoo; see, *e.g.*, Bryzgalova, Huang, and Julliard (2019), Feng, Giglio, and Xiu, 2020, Freyberger, Neuhierl, and Weber, 2020, and Kozak, Nagel, and Santosh, 2020. However, the many approaches have led to different sets of “correct” factors; these sets appear to depend on the sample period, how one adjusts for implementation costs, and what techniques are used to establish statistical significance. A general finding is that there are far fewer significant factors in reality than have been proposed.

In parallel, the financial econometrics literature has extended CAPM by using high-frequency data in a continuous-time setting to improve the estimation and identification of systematic risk. For instance, a large number of papers have shown that market jumps make up an important component of the volatility of the market portfolio (Barndorff-Nielsen and Shephard, 2005; Huang and Tauchen, 2005; Lee and Mykland, 2008; Andersen, Bollerslev, and Diebold, 2007). Consequently, recent work has addressed jumps by developing econometric techniques to estimate continuous and jump betas separately and over time. Todorov and Bollerslev (2010) estimate a single-factor model with differing continuous and jump betas, while Bollerslev, Patton, and Quaadvlieg (2016) extend these results. Some of this literature has also used options data to study jump risk; see, for example, Andersen, Fusari, and Todorov, 2015, Bollerslev and Todorov, 2011a, Bollerslev, Todorov, and Xu, 2015, and Broadie, Chernov, and Johannes, 2007. All of these papers have generally found that the market jump risk premium is fairly large.

In addition to jumps, recent work has also focused on decomposing variances into semi(co)variances and jumps into semijumps (positive/negative jumps). While the basic

² This is one of many approaches that the asset pricing literature has taken. For a full review, see Cochrane (2017).

idea to split variances into semivariances goes back to Roy (1952) and Markowitz (1959), more recent work has reapplied it to high-frequency data, which is particularly useful for estimating second moments and differentiating the sign of returns. Bollerslev (2021) provides a review of the semi(co)variation and semijump literature. In general, downside variation and negative jumps are particularly important for understanding financial market volatility and asset returns. Similarly, Bollerslev, Patton, and Quaadvlieg (2020) use semi(co)variances to decompose factor betas into four semibetas and find that the semi-beta associated with negative market/asset returns predicts significantly higher returns relative to the other betas.

My paper connects the all of these literatures. As in the financial econometrics literature, I use high-frequency data in a continuous-time setting to estimate a sufficiently complex model of multifactor risk by separately studying continuous and (semi)jump returns. Then, as in the asset pricing literature, I use these estimates of factor risk, the continuous and (semi)jump factor betas, to explain the cross-section of expected returns. My work is closest to that by Aït-Sahalia, Jacod, and Xiu (2021), who introduce the Continuous-Time Fama-MacBeth regression to estimate continuous and jump risk premia in a multifactor setting. I use the same estimator to show that semijump risk premia is a major component of risk premia for many factors in the zoo. Additionally, through my decomposition of expected returns, I show that jump risk premia also appears to explain the majority of cross-sectional variation in expected returns. These results tie together the large body of high-frequency work on jumps with the multifactor models commonly used in asset pricing.

The outline for the rest of the paper is as follows. First, in Section 2, I describe the construction of my high-frequency factors and report their descriptive statistics. Then, in Section 3, I introduce my model and econometric methodology. In Section 4, I report my risk premia estimates for the workhorse models and my factor zoo. In Section 5, I introduce my risk premia decomposition and report the associated results. Finally, in Section 6, I conclude.

2 Data

2.1 High-Frequency Prices

In order to construct high-frequency portfolios, I first construct a dataset of high-frequency prices for individual stocks. Recall that characteristic-sorted portfolios are traditionally formed by averaging the returns on thousands of stocks to isolate factor risk. Hence, it is necessary to collect data on all stocks regardless of their market cap or volume. This deviates from most of the high-frequency financial econometrics literature, which focuses on stocks with minimal microstructure noise. However, this noise will be effectively diversified away in the value-weighted portfolio returns. Thus, factor portfolios can be studied at a high-frequency in the same way as large-cap and mid-cap stocks.

I begin by obtaining prices from January 1996 to December 2020 for all common stocks listed on the three primary exchanges (NYSE, NASDAQ, NYSEMKT). This universe of stocks is based on Fama and French (1993). Next, I obtain high-frequency prices

from the NYSE Trade and Quote Database (TAQ) through WRDS; these prices are cleaned using standard procedures (Barndorff-Nielsen et al., 2008) and sampled at a 5-minute frequency from 09:30 to 16:00.³ I combine the high-frequency TAQ prices with open/close prices and adjusted returns from CRSP; these adjusted returns handle dividends, stock splits, delisting returns, and other events. Lastly, I construct high-frequency value weights using the same procedure as Aït-Sahalia, Kalnina, and Xiu (2020) – these are used later on for the factor portfolio returns. Overall, my high-frequency dataset of prices/returns consists of 14,610 stocks, 2.38 billion high-frequency return observations, and about 163,082 observations per stock.⁴

2.2 Factors and Signals

I combine Chen and Zimmermann (2020) and Jensen, Kelly, and Pedersen (2021), henceforth CZ and JKP, to produce a set of 268 characteristic factors based on 151 different papers.⁵ This set is fairly comprehensive given that CZ and JKP themselves build on previous literature reviews done by Harvey, Liu, and Zhu (2016), Green, Hand, and Zhang (2017), and Hou, Xue, and Zhang (2020). For each of the factors, I obtain the associated signals from data and code provided by JKP and CZ. But, some factors do not have signal data that covers my sample, 1996-2020, while others have duplicate signal data because they appear in both in both JKP and CZ. To solve the first issue, I drop all factors with missing data and, to solve the second, I opt for JKP signal data over CZ signal data and drop the duplicated factors. After pruning, I am left with 218 factors based on 153 signals from JKP and 65 signals from CZ.

In addition to the 218 factors from the literature, I also consider the workhorse factors and industry portfolios. The workhorse factors are MktRf, SmallMinusBig, HighMinusLow, RobustMinusWeak, and ConservativeMinusAggressive plus Momentum; they are based on market return, market capitalization, book-to-market values, operating profitability, investment, and past returns (Fama & French, 2015). The industry portfolios are based on 48 industry classifications.⁶

In total, my “high-frequency factor zoo” consists of 272 portfolios: 218 characteristic-sorted factors from the literature (JKP+CZ), 6 factors from Fama and French (2015), and 48 industry portfolios.

2.3 High-Frequency Factor Portfolios

A high-frequency factor portfolio is essentially a traditional factor portfolio but simply observed at a high-frequency. Such portfolios can be constructed using the same approach as traditional, low-frequency portfolios but using high-frequency prices/returns instead.

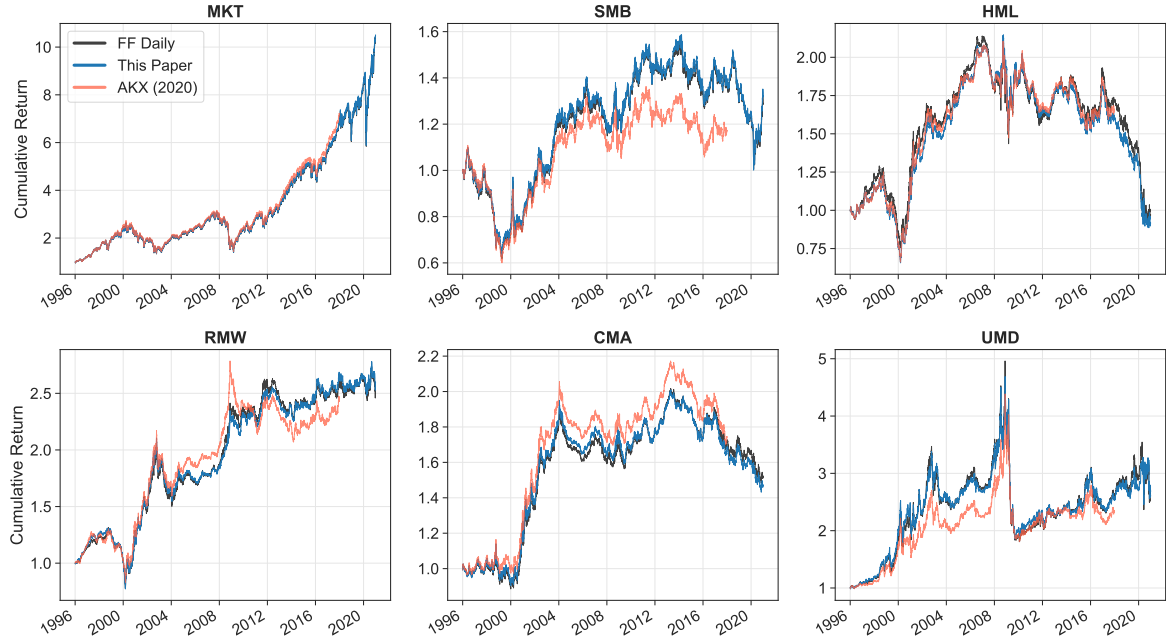
³ When constructing the portfolios and performing regressions, I will aggregate the data up to 15-minutes based on the signature plots shown in Section A.3. This resampling is done to minimize microstructure noise while maintaining an adequate amount of data.

⁴ I describe my data cleaning methodology in greater detail in the Online Appendix.

⁵ The full list of factors and their respective citations are given in the Online Appendix.

⁶ These industry classifications obtained from French’s website (also see Fama and French, 1997). Additionally, I also use French’s website to obtain risk-free rates. These rates are daily but high-frequency data for the risk-free asset will not be necessary in what follows.

Figure 1: CUMULATIVE RETURNS ON THE FAMA-FRENCH 5+1 FACTORS



Note: I plot cumulative returns for each of the Fama-French 5+1 factors based on three different data sources. The black line labelled FF Daily refers to the daily returns obtained from French’s website. The dashed blue line is based on my own replication of the factor. And, the light orange line is based on cumulative returns for the high-frequency portfolios from Aït-Sahalia, Kalnina, and Xiu (2020) which are available on Xiu’s website. All returns are aggregated to daily for visibility.

In order to construct high-frequency portfolios, I use two different methodologies. Firstly, I use a simple and uniform methodology for the 218 JKP+CZ (2021, 2020) factors and the 48 industry portfolios. And secondly, I use a methodology faithful to the original construction for the Fama-French (2015) 6 factors. For all 272 portfolios, I limit the universe of stocks to the Fama and French (1993) universe described earlier: common stocks on NYSE/NASDAQ/NYSEMKT. I describe my portfolio construction methodology in greater detail in the Appendix Section A.1.

To illustrate the accuracy of my high-frequency FF6 factors, I plot their cumulative returns in Figure 1. My own high-frequency portfolio returns, aggregated to daily, are shown in blue, and the original portfolio returns, obtained from French’s website, are shown in black. In addition, I include portfolio returns from Aït-Sahalia, Kalnina, and Xiu (2020) who also produce high-frequency versions of the Fama-French 5+1 factors. The R^2 between the *cumulative* returns for my daily-aggregated high-frequency factors and the original Fama-French daily factors are over 99% for all six factors. The R^2 for the daily simple returns for Mkt, SMB, HML, RMW, CMA, and UMD are 99.9%, 99.6%, 96.8%, 96.7%, 99.1%, and 96.4%. The R^2 values are not perfect but they only marginally deviate from 100%, likely due to idiosyncratic decisions made when cleaning the data.

Lastly, to alleviate concerns over microstructure noise, I rely on 15-minute returns throughout the paper. I discuss this decision more thoroughly in the Appendix (Section A.3) where I also provide signature plots for my factors.

Table 1: FACTOR ZOO – RISK PREMIA AND ALPHAS

Variable	Sig. Factors		$Avg(R^2)$	Pricing Errors	
	Standard	MHT		$Avg(\alpha)$	$\frac{Avg(\alpha)}{Avg(Return)}$
Raw Returns	28	2	0.00%	1.12%	100.00%
CAPM Alphas	62	4	8.10%	1.82%	163.55%
FF3 Alphas	84	32	26.93%	1.87%	167.83%
FF5 Alphas	59	8	36.26%	0.77%	68.75%
FF6 Alphas	54	5	40.50%	0.41%	37.20%

Note: Each entry represents an annualized estimate of the risk premia or the alpha for a particular high-frequency portfolio from the set of JKP+CZ factors. The next two columns report the number of factors that pass a 5% level test; the “Standard” column is based on the $|t| > 1.96$ rule while the “MHT” column also applies a Benjamini and Yekutieli (2001) correction for multiple hypothesis testing. The average R^2 s are based on those from the time-series regressions used to estimate the alphas for each factor. The $Avg(\alpha)$ column reports the average alpha while the last column divides the average alpha estimates by the average return for the cross-section of factors. The estimates employ 15-minute returns for the full 25-year sample of 1996-2020. The underlying factor portfolios are long in the direction specified by the literature.

2.4 Descriptive Statistics

In this section, I make three general points about the factor zoo. Firstly, the standard risk premia and alphas are not particularly large. This point is in agreement with several papers that argue that many factors in the zoo underperform when taken out-of-sample (see, e.g., Feng, Giglio, and Xiu (2020), Harvey, Liu, and Zhu (2016), Hou, Xue, and Zhang (2020), Jacobs and Müller (2020), and McLean and Pontiff (2016)). Secondly, based on the structure of the correlations between factors, the zoo appears to lie in a relatively low dimensional space. In other words, many of the factors are strongly correlated with one another, implying that much of the zoo is redundant. Motivated by this finding, I group the factors into clusters based on Jensen, Kelly, and Pedersen (2021), which I use to simplify the analysis in the sequel when I estimate continuous and jump risk premia. And, thirdly, there is clear evidence that the high-frequency factor portfolios have a non-trivial jump component. Furthermore, these factor jumps appear to be orthogonal to the market and co-occur with jumps in a sample of SP500 stocks – this finding consistent with the idea that the factors themselves can be used to capture non-market, systematic jump risk.

2.4.1 Expected Returns

I begin by discussing the risk premia and the alphas for the 218 JKP+CZ factors. I estimate the risk premia by averaging the raw returns and estimate the alphas using time-series regressions. For the regressions, I consider four specifications: CAPM, FF3, FF5, and FF6. I report my results for the 218 factors in Table 1.

Unsurprisingly, many of the factors do not draw a statistically significant risk premia. That is, of the 218, only 28 have a statistically significant ($|t| > 1.96$) risk premia, while just 2 survive a multiple hypothesis testing correction (Benjamini & Yekutieli, 2001).

These results are similar to those from Hou, Xue, and Zhang (2020) who find that many factors in the zoo cannot pass a simple t-test. In contrast, the alphas generally do show statistical significance. In fact, about twice as many factors have statistically significant CAPM alphas than risk premia. This finding is consistent with the idea that factors capture systematic risk left unpriced by workhorse models. Interestingly, the largest number of significant alphas is obtained under the FF3 specification. This finding is also not surprising given that FF-MKT, FF-SMB, and FF-HML were invented quite early relative to the rest of the factor literature (Fama & French, 1993). On the other hand, the remaining factors in the FF6 are more recent and were intentionally introduced to help explain the many so-called anomalies proposed after the popularization of the Fama-French 3 factor model (Fama & French, 2015).

For the same reason, these later three FF6 factors – Investment, Profitability, and Momentum – appear to have economically significant pricing implications. That is, when moving from an FF3 specification to an FF6 specification, the resulting average alphas decline from 1.87% to just 0.43%. Likewise, the average alphas relative to the total return (the last column in Table 1) decline from 167% to only 37%. A plot of the alphas under each specification, Figure A.4 in the Appendix, also highlights the explanatory power of these latter three FF6 factors. In a similar vein, ability to explain time-variation, measured by the average time-series R² reported in the third column of Table 1, tells the same story: CAPM explains a relatively small portion of time-variation variation in the factor zoo, while the FF3 and FF5 models explain notably more.

To sum up, the factors appear to draw little risk premia but substantially more alpha as computed under the FF3 and CAPM specifications. The newer half of the FF6 factors explain much of this alpha, along with a substantial amount of the time-variation within each of the portfolios. But, in any case, the risk premium of an asset says little about its continuous and jump risk premia. Hence, studying these more granular risk premia remains an interesting pursuit regardless of the findings reported above.

2.4.2 Covariation and Clusters

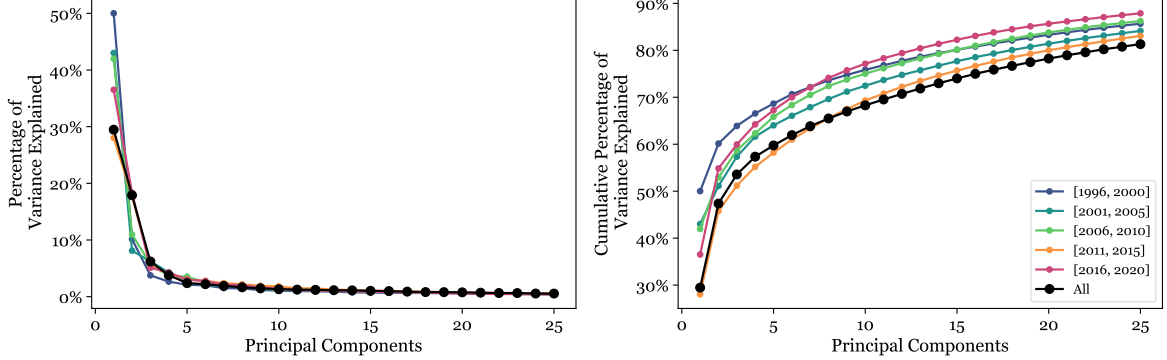
I now turn to the covariation between the JKP+CZ factors. Based on the literature, it is well known that many of the proposed factors lie in a lower dimensional space. To check whether this holds for my set of portfolios, I run a principal components analysis (PCA) on their returns. I also repeat the analysis on a five-year basis to check for time variation in the statistical factor structure of the zoo.

I plot my results in Figure 2 which reports the individual and cumulative variation explained by each principal component (PC). As expected, a few PCs explain a significant portion of the overall variation. The first five, ten, and eleven principal components cumulatively explain about 50%, 60%, and 70% of the overall variation, respectively. Interestingly, re-estimating PCA on each five-year subsample in my dataset produces substantially higher explanatory power. A five PC model rerun on a five-year basis explains, on average across years, 65% of overall variation. In other words, the 218 factors lie in a lower dimensional space and this space seems to be time-varying.

Going a step further, in the spirit of Pelger (2019) and Pelger (2020), I run separate

principal component analyses on the jump and continuous returns (see also Ait-Sahalia and Xiu (2015) and Ait-Sahalia and Xiu (2017)). To do so, I separate the continuous and jump returns using the standard thresholding procedure (detailed in the following Section 2.4.3) and rerun separate analyses on the two sets of returns. I plot my results in Figure A.5. The variation measures for the continuous and jump returns are somewhat lower but still in agreement with the idea that the first few statistical factors explain a disproportionate amount of the total variation.

Figure 2: FACTOR ZOO – PRINCIPAL COMPONENTS ANALYSIS



Note: I run a principal components analysis on the high-frequency returns of my 218 JKP+CZ portfolios over each five year subsample in my 25-year dataset. I also include results for a PCA run on the full sample and denoted “All.” The first subplot reports the portion of total variation explained by each principal component up to 25 in total. The second subplot is similar but reports the cumulative total variation explained.

Motivated by these findings, I assign my factors to “clusters” to reduce the dimensionality of my zoo while preserving economic interpretability and simplifying the analysis that follows. To define the clusters, I rely on the thirteen factor classifications introduced by Jensen, Kelly, and Pedersen (2021).⁷ In doing so, I (i) avoid contributing to the “zoo” of factor classifications by producing my own clusters, (ii) remain consistent with past work, and (iii) can readily cluster my 153 JKP-based portfolios by reusing the assignments from their paper.

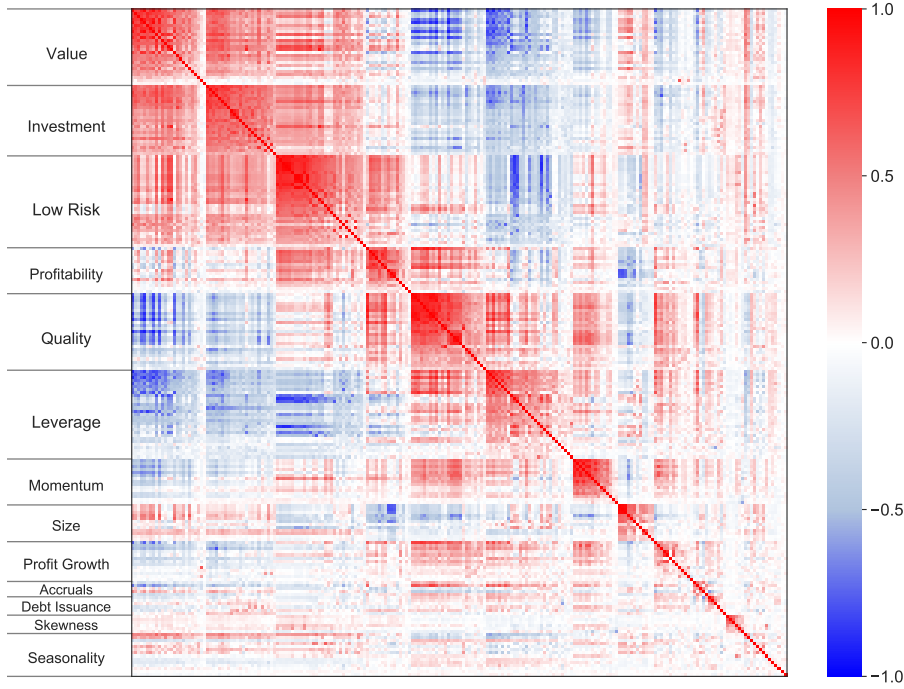
For my remaining 65 factor portfolios based on Chen and Zimmermann (2020), I need to manually assign classifications. I do so by assigning each unclassified portfolio to one of the thirteen JKP clusters. To detail my procedure, let $\rho(k_{CZ}, k_{JKP})$ be the correlation between the CAPM residuals⁸ of some CZ-based factor portfolio, k_{CZ} , and a JKP-based factor portfolio, k_{JKP} . Furthermore, let $\mathcal{C}$ be the set of sets of JKP factors, grouped by classification. The classification assigned to CZ portfolio k_{CZ} is given by

$$\text{Classification}(k_{CZ}) = \arg \max_{C \in \mathcal{C}} \left(\frac{1}{\#\mathcal{C}} \sum_{k_{JKP} \in C} \rho(k_{CZ}, k_{JKP}) \right).$$

⁷ Jensen, Kelly, and Pedersen group factors together using hierarchical agglomerative clustering from Murtagh and Legendre (2014). They cluster on correlations and use the ward linkage criterion. Their estimated correlations between factor portfolios are based on CAPM residuals, which are estimated using monthly data.

⁸ I estimate the residuals by regressing the portfolios against my high-frequency Fama-French market factor, using the full-sample of 15-minute and overnight returns from 1996-2020.

Figure 3: CORRELATIONS BETWEEN THE CAPM RESIDUALS OF THE JKP+CZ FACTORS



Note: I plot the correlations between the CAPM residuals of the 218 JKP+CZ factors. The residuals are computed using simple regressions on the high-frequency Fama-French market factor. The left-hand-side of the graph shows the cluster assignments for each of the factors; the labels for the x-axis follow the same ordering as that of the y-axis. The underlying clusters are based on Jensen, Kelly, and Pedersen (2021). The sample period is the full sample, 1996-2020.

In other words, I compute a correlation measure between each CZ factor and the set of 13 JKP clusters. I then assign the CZ factor to the cluster with which it has the highest correlation. This methodology is a natural way to append the CZ factors, because the Jensen, Kelly, and Pedersen (2021) classifications were originally produced by clustering on correlations between CAPM residuals. And, this methodology avoids entirely redoing the clustering, thereby maintaining consistency with the assignments from Jensen, Kelly, and Pedersen (2021).

In the Online Appendix, I report each CZ factor's cluster assignment and their average correlation with the factors in their assigned cluster. The CZ portfolios have an average correlation ρ of 35% with respect to the JKP portfolios in their assigned clusters. This number is fairly large in magnitude, since the average absolute correlation between all 218 factors is 11%.

To visualize the covariance structure and the clusters, I produce a heatmap of the realized correlations between the CAPM residuals of my 218 JKP+CZ factors in Figure 3. Within each cluster, factors are ordered by the average correlation measure described above. Note that there is a clear block diagonal structure, suggesting that the clustering approach effectively capturing groups of related factors. And, although some factors have relatively weak correlations with all of the other factors, most appear to fit clearly into a cluster. Interestingly, some of the clusters also appear to form larger, moderately correlated blocks. However, the between-cluster correlations for these broader blocks

appear to be too weak to reasonably perform further merges.

To better corroborate this point, I also study the time-variation in the covariance structure by reproducing [Figure 3](#) on a yearly basis. The resulting figures are provided in the Online Appendix. In short, I find that between-cluster correlations change significantly over time, while the within-cluster correlations generally remain stable. This finding suggests the clusters are robust to time-variation in the statistical factor structure underlying the zoo and, more importantly, implies that merging any of the above clusters would oversimplify the classifications. All in all, the clustering approach appears to effectively separate groups of factors without being too coarse or too fine.⁹

2.4.3 Jumps & Co-Jumps

A critical advantage of using high-frequency return data is the ability to identify “jumps,” discontinuous moves in a semimartingale process, with reasonable statistical precision. In this section, I describe my methodology for identifying jumps and provide some simple summary statistics regarding the number of jumps and the level of jump variation in my factors. Additionally, since factor jumps must be systematic to have non-trivial pricing implications, I further test for co-jumps between my factors and a set of SP500 stocks using the methodology of Jacod and Todorov ([2009](#)).

To start, I describe my jump detection procedure. I rely on the standard thresholding approach proposed by Mancini ([2001](#)). Put simply, this approach works by classifying return a jump if its absolute value exceeds a given threshold. Asymptotically, the threshold is assumed to shrink towards zero at a rate slower than the sampling frequency $\sqrt{\Delta_n}$; this makes it possible to perfectly separate jumps, which are of fixed size, from Brownian motion, which scales with the square root of the sampling frequency. To implement this approach in a finite-sample setting, I follow Barndorff-Nielsen and Shephard ([2006](#)), who suggest sizing the threshold using bipower variation, and Bollerslev, Todorov, and Li ([2013](#)), who suggest adjusting for intraday seasonality in volatility using a time-of-day correction.

Formally, the jump detection thresholds at time i for stocks (indexed by m) and factors (indexed by k) are computed as

$$u_n^s(i, m) \equiv \alpha \sqrt{\tau_{\ell(i),m}^s BV_{d(i),m}^s} \Delta_n^\varpi \quad (1)$$

$$u_n^f(i, k) \equiv \alpha \sqrt{\tau_{\ell(i),k}^f BV_{d(i),k}^f} \Delta_n^\varpi \quad (2)$$

where α is a tuning parameter, τ is a time-of-day adjustment, $\ell(i)$ is the time-of-day for return i , BV is bipower variation, $d(i)$ is the day for return i , and ϖ is another tuning parameter. I describe how I compute the bipower variation and the time-of-day correction, along with how I classify the overnight returns, in the Online Appendix. With respect to the tuning parameters, I use $\alpha = 3.0$ and $\varpi = 0.49$ for my main specification but vary these parameters in my robustness checks. To provide some intuition, note

⁹ I produce equivalent covariance heatmaps for the identified jump and continuous returns as well. These are provided in the Online Appendix and look quite similar to the simpler [Figure 3](#) – in other words, the clusters also “hold” for the continuous and jump returns.

that $\sqrt{\tau_{\ell(i),k}^f BV_{d(i),k}^f}$ is an estimate of the volatility of the Brownian motion component of factor k at time i , while $\Delta_n^\varpi \approx \Delta_n^{0.5}$ scales this volatility to the sampling frequency of the returns. Hence, in finite sample, this means that a return is classified as a jump if it exceeds $\alpha = 3.0$ standard deviations in magnitude.

Next, I use this procedure to classify jumps in my high-frequency factors. To help condense the analysis that follows, I focus on thirteen “cluster portfolios,” which I assemble using the first principal component of the factor returns within each of the thirteen clusters defined earlier.¹⁰ By construction, these portfolios capture the shared variation within each cluster and thus facilitate a lower-dimensional but readily interpretable analysis of the zoo. To substantiate this point, I plot the cumulative returns for each of the cluster portfolios and their underlying factors in [Figure A.6](#). In general, the cluster portfolios match the variation and average returns of the factor clusters they represent.

Following the plan of summarizing the zoo through the cluster portfolios, I identify the jump returns therein and report my results in [Table 2](#). The first column shows the average number of intradaily jumps per year for each of the cluster portfolios plus those in my high-frequency market factor. For all portfolios, this statistic takes similar values, hovering around 60 jumps per year. The average number of days with at least one jump is slightly lower but consistent across the portfolios, being around 50 per year. While these number may appear large in magnitude, they are consistent with similarly computed counts from Bollerslev, Law, and Tauchen (2008) (see Table 1 of their paper). Additionally, although it may seem prudent to further reduce Type I error by using a more conservative jump truncation threshold α ([Equation 2](#)), the seemingly large number of jumps will also help ensure a more reasonable bias-variance trade-off when performing jump regressions in the sequel.¹¹

To check if the factors are simply cojumping with the market, I also count the subset of jumps in each of the portfolios that do not co-occur with jumps in the market portfolio. For simplicity, I estimate this subset by simply filtering out intervals with detected jumps in each of the portfolios that also contain a detected jump in the market. I report the yearly averages of my estimated non-market jump counts in the second column of [Table 2](#). I find that the vast majority of jumps in the cluster portfolios do not co-occur with the market, consistent with the fact that the underlying characteristic-sorted factors are also uncorrelated with the market.

The next two columns in [Table 2](#) further corroborate the existence of jumps in the cluster portfolios. The “RJV” column reports the fraction of intraday variation that stems from jumps. This statistic is computed by normalizing the difference between realized and bipower variation by realized variation itself (see Huang and Tauchen, 2005). Since realized variation captures the total quadratic variation while bipower variation captures

¹⁰ Since principal components are simple linear combinations of the underlying data, the portfolios are well-defined. Additionally, in order to keep each cluster portfolio’s weights (eigenvectors) reasonable, I force the sum of the weights to add up to one. This adjustment serves as a normalization to keep the scale of the portfolios economically reasonable, although the choice of total weight is fundamentally meaningless since the underlying characteristic-sorted portfolios are zero-investment. I provide additional details regarding the portfolio weights in the Online Appendix.

¹¹ If one were to set $\alpha = 3.5$, the average number of jumps per year, across all fourteen portfolios, would lie around 15.

Table 2: CLUSTER PORTFOLIOS – JUMP STATISTICS

Portfolio	Average Jumps per Year		RJV	z(RJV)
	All	Non-Market		
Market	56.12	0.00	0.0374	5.088
Accruals	56.72	52.56	0.0671	15.073
Debt Issuance	56.48	52.68	0.0709	11.983
Investment	61.68	56.08	0.0573	5.742
Leverage	66.00	58.72	0.0435	1.862
Low Risk	60.68	50.92	0.0400	3.433
Momentum	58.80	52.12	0.0660	7.492
Profit Growth	56.56	50.96	0.0671	11.037
Profitability	63.28	57.00	0.0534	8.011
Quality	60.84	55.20	0.0645	10.829
Seasonality	55.64	46.36	0.0652	5.265
Size	55.40	49.40	0.0463	7.346
Skewness	62.76	57.00	0.0705	8.341
Value	66.00	59.20	0.0567	4.497

Note: For each of the cluster portfolios, I report the average number of jumps per year, the average number of non-market jumps per year, the relative jump variation, and the z-statistic for the full sample relative jump variation. The first statistic is computed as the yearly average of number of the identified intradaily jumps in each of the portfolios. The second statistic is computed similarly but discloses jumps in the portfolios that co-occur with those in the market. The third statistic, the relative jump measure, is computed as the yearly average of the statistic $(RV - BV)/RV$ where RV is the realized variance and BV is the bipower variance. The last statistic is computed as $RJV/\sqrt{(v_{bb} - v_{qq})\frac{1}{M}\frac{TP}{BV^2}}$ where $v_{qq} = 2$, $v_{bb} \approx 2.609$, M is the number of return observations in the sample, TP is the realized tripower quarticity, and BV is the bipower variation. The tripower quarticity is computed as $TP = M(0.8309)^{-3}\frac{M}{M-2}\sum_{j=3}^M |r_{j-2}|^{4/3}|r_{j-1}|^{4/3}|r_j|^{4/3}$ where r_j refers to the j 'th return in the sample (Barndorff-Nielsen & Shephard, 2004). All four statistics are based on intradaily 15-minute returns across the full sample, 1996-2020.

only the continuous component of quadratic variation, the positive RJV values suggest that a non-trivial fraction of the overall variation in the cluster portfolios comes from jumps. To formally test whether the RJV values are statistically significantly greater than zero, I compute a test statistic for RJV based on Barndorff-Nielsen and Shephard (2004). Assuming the sampling frequency grows arbitrarily large, this statistic is asymptotically normal under the null but grows arbitrarily, positively large under the alternative. I compute the statistic on my full sample of intraday returns and report my results in the last column of Table 2. Under a 5% significance level, I can reject the null of zero jump variation for all fourteen portfolios. Going a step further and rerunning test on a daily basis, I reject the null at a 5% level for approximately 12% of the days in my sample for each of the fourteen portfolios. This finding is consistent with equivalent estimates for the SP500 index from Huang and Tauchen (2005) and is approximately consistent with the average number of days with jumps mentioned earlier.

Overall, these results lend further support to the idea that there exists non-market,

non-diversifiable jump risk. That is, if non-market jumps were diversifiable, the factor portfolios, constructed by averaging the returns on thousands of stocks, should be conditionally Gaussian. However, the cluster portfolios have occasionally, excessively large returns (detected jumps) and exhibit jump variation that is both economically and statistically as significant the jump variation in the market portfolio. All together, these results imply that there exist non-market jumps that do not “wash out” under aggregation.

To provide more straightforward evidence that the high-frequency factors do indeed capture systematic non-market jumps, I now perform direct tests for cojumps between my factors (proxied by the cluster portfolios) and a sample of SP500 stocks using a test statistic proposed by Jacod and Todorov (2009). In short, this statistic is essentially a higher order power variation of a bivariate semimartingale process and can be used to test the null of disjoint jumps against the alternative of at least one cojump for a pair of assets. The asymptotic distribution and computation of the statistic is somewhat complex, so I defer a full exposition and the implementation details to the Online Appendix. The test itself also presents a simple way to falsify the claim that the factor jumps are systematic. That is, if I support the null of no cojumps across a broad set of assets, it would imply that the factor jumps are diversifiable and their pricing implications immaterial.

In any case, I compute the test statistic on a monthly basis using the intraday 15-minute returns and report the fraction of rejections ($p < 0.05$) in Table A.2. Overall, I find broad rejections of the null across all pairs of factors and stocks. More precisely, across all the factor-stock pairs, about 90% of the months in my sample contain at least one cojump. Additionally, combined with the earlier finding that many of the jumps in the cluster portfolios do not co-occur with the market, my results here further imply a non-rejection of the idea that there exist systematic, market-neutral jumps in stock prices (see also Lin and Todorov (2019) and Jacod, Todorov, and Lin (2022)). A natural follow-up question is whether these jumps have non-trivial pricing implications. To investigate this, I now proceed to my methodology for estimating the continuous and jump risk premia of the factors.

3 Methodology

Since the focus of this paper is on continuous and jump risk premia, I first set some notation to better delineate these concepts. To start, note that the stochastic discount factor (SDF) may be written in continuous-time as

$$M_t = e^{-\int_0^t r_s^f ds} \cdot \mathcal{E} \left(- \int_0^t \tilde{\lambda}_s^C dW_s + \int_0^t \int_{\mathbb{R}} (\tilde{\lambda}_{s,z}^J - 1) \tilde{\mu}(ds, dz) \right), \quad (3)$$

where r_s^f is the risk-free rate, $\mathcal{E}(\cdot)$ is the stochastic exponential, and $(\tilde{\lambda}^C, \tilde{\lambda}^J)$ are terms that price continuous and jump risk (see, e.g., Bollerslev, Patton, and Quaedvlieg, 2016; Duffie, Pan, and Singleton, 2000; Ho, Perraudin, and Sørensen, 1996). As usual, covariation with this factor determines an asset’s risk premium.

In a continuous-time setting, we may go a step further and decompose the covariation with the pricing kernel into two separate components: quadratic covariation with

the continuous component and quadratic covariation with the jump component.¹² Correspondingly, any spot risk premia can be readily decomposed into compensation for continuous risk and compensation for discontinuous risk. More formally, we may write

$$\mu_t - r_t^f = \beta_t^C \lambda_t^C + \beta_t^J \lambda_t^J, \quad (4)$$

where $\mu_t - r_t^f$ is the expected excess spot return of the asset, (β_t^C, β_t^J) are the usual continuous and jump betas between the asset return and the SDF, and $(\lambda_t^C, \lambda_t^J)$ are the continuous and jump risk premia of the SDF.¹³

The structure of the excess returns is reminiscent of a standard factor model. However, unlike traditional discrete-time models, this setup clearly differentiates the risk premium from exposure to systematic diffusive risk from the premium from exposure to systematic jump risk. This observation raises to the following questions: (i) do continuous and jump risk premia differ; (ii) are semijump risk premia nontrivial; and (iii) are jumps useful for explaining the cross-section of returns? To answer these questions, I estimate the continuous and jump risk premia for each factor in my dataset using Continuous-Time Fama-MacBeth regressions (Aït-Sahalia, Jacod, & Xiu, 2021). For completeness, I detail this regression below.

3.1 Model Setup

All stochastic processes are defined on some filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \geq 0}, \mathbb{P})$. The set of $k \times m$ matrices is given by $\mathcal{M}_{k,m}$, while the set of PSD $k \times k$ matrices is given by $\mathcal{M}_k^+$. The smallest eigenvalue of a matrix A is given by $\zeta(A)$. The notation ΔX_t for some arbitrary stochastic process X at time t refers to $X_t - X_{t-}$. Additionally, the i 'th time interval is denoted by $I_i^n = ((i-1)\Delta_n, i\Delta_n]$ and the i 'th return is denoted $\Delta_i^n X = X_{i\Delta_n} - X_{(i-1)\Delta_n}$.

To begin, we observe a set factors which jointly follow a K -dimensional process:

$$F_t = F_0 + \int_0^t \mu_s^F ds + F_t^C + F_t^J \quad (5)$$

$$F_t^C = \int_0^t \sigma_s^F dW_s^F \quad (6)$$

$$F_t^J = \int_0^t \int_{\mathbb{R}^K} \delta^F(s, z) \underline{p}^F(ds, dz) \quad (7)$$

where W^F is a Brownian motion and $\underline{p}^F$ is a Poisson random measure on $\mathbb{R}_+ \times \mathbb{R}^K$ with

¹² Recall that the quadratic covariation between a purely continuous and a purely discontinuous process is zero. Consequently, since the pricing kernel consists of a continuous and discontinuous component, it is possible to cleanly decompose quadratic covariation between the kernel and some asset return into two components: the continuous part of quadratic covariation and the discontinuous part of quadratic covariation. See Back (1991) for a more formal discussion.

¹³ This equation is similar to equation (7) in Bollerslev, Patton, and Quaedvlieg (2016). The general point about decomposing returns is also discussed in Bollerslev and Todorov (2011a) and more formally proven in Back (1991). If returns are based on log-prices, then there also needs to be a convexity adjustment.

the intensity measure $\underline{q}(dt, dz) = dt \otimes \nu(dz)$ with ν being a σ -finite measure on $\mathbb{R}^K$. The process F_t^C represents the continuous component of the factors, while F_t^J represents the jump component.

The jump component for each factor is allowed to represent multiple sources of jump risk. For instance, positive jumps in a factor may demand a different risk premium than negative jumps. Accounting for this heterogeneity requires some additional notation. For instance, for a “semijump” model, we may write

$$\tilde{F}^k = F^{C,k} \quad (8)$$

$$\bar{F}^{J,k,Pos} = \sum_{s \leq t} \Delta F_s^k \cdot \mathbb{1}_{[\Delta F_s^k > 0]} \quad (9)$$

$$\bar{F}^{J,k,Neg} = \sum_{s \leq t} \Delta F_s^k \cdot \mathbb{1}_{[\Delta F_s^k < 0]}. \quad (10)$$

Here, the K dimensional vector $\tilde{F}$ captures factor continuous risk, while the $H = 2 \cdot K$ dimensional vector $\bar{F}^J$, obtained by stacking the positive and negative factor jumps, captures factor semijump risk. These $K + H = 3K$ new “risk factors” provide a more granular representation of systematic risk than afforded by both standard discrete-time factor models and continuous-time factor models that treat all jumps homogeneously. In contrast, a simpler but less granular alternative is a specification where the factors are split into two components, continuous and jump, for a total of $K + H = 2K$ risk factors. A final, trivial option is to further combine the continuous and jump components for each factor into a single risk factor, echoing a traditional factor model.

The next step is to define the processes for asset prices. I assume the following spot-linear factor structure:

$$P_t = P_0 + \int_0^t \beta_s^C d\tilde{F}_s + \int_0^t \beta_s^J d\bar{F}_s + P_t^I \quad (11)$$

$$P_t^0 = \int_0^t r_s^f ds \quad (12)$$

$$P_t^I = \int_0^t \mu_s^I ds + \int_0^t \sigma_s^I dW_s^I + \int_0^t \int_E \delta^I(s, z) \underline{p}^I(ds, dz). \quad (13)$$

Here, P_t is a M -dimensional vector of log prices, r_t^f is an optional processes representing the risk-free rate, β_t^C is a predictable $\mathcal{M}_{M,K}$ process, and β_t^J is a predictable $\mathcal{M}_{M,H}$ process. The idiosyncratic component P_t^I is defined with an M -dimensional Brownian motion W^I and a Poisson random measure $\underline{p}^I$ with intensity $\underline{q}$; each of its characteristics $\mu^I \in \mathbb{R}$, $\sigma^I \in \mathcal{M}_{M,M}$, and $\delta^I \in \mathbb{R}^M$ are predictable.

To isolate the excess returns, we may rewrite the price process as,

$$P_t = P_0 + \underbrace{\int_0^t (\mu_s - r_s^f) ds}_{\text{Excess Return}} + \underbrace{\int_0^t \beta_s^C d\tilde{F}_s^* + \int_0^t \beta_s^J d\bar{F}_s^*}_{\text{Martingale Terms}} + P_t^I, \quad (14)$$

where $\tilde{F}^*$ and $\bar{F}^*$ are compensated versions of the original processes. Under a factor

structure for the SDF and a no-arbitrage condition, excess returns follow

$$\mu_t - r_t^f = \beta_t \lambda_t = \beta_t^C \lambda_t^C + \beta_t^J \lambda_t^J, \quad (15)$$

for all t (Aït-Sahalia, Jacod, & Xiu, 2021). This equation directly mirrors the simpler Equation 4 and may be seen as an extension of the APT arguments in Ross (1976) to a continuous-time setting. The Appendix provides further details concerning the necessary, more technical assumptions involved.

A few remarks are in order. Firstly, Equation 12 imposes a spot linear factor structure on the asset returns. This assumption is not as restrictive as it appears, because the continuous betas β_t^C are allowed to vary over time thus capturing non-linearities between asset prices and the factors $\tilde{F}$. However, the jump betas β^J are not allowed to vary arbitrarily over time. This restriction arises because it is not possible to identify the jump beta β_t^J at time t if a jump does not occur at that time. Hence, in practice, I perform yearly rolling regressions as a heuristic solution.¹⁴

Secondly, it should also be noted that individual stocks may drop in and out of the sample, which would violate the implicit assumption that we can observe prices for all assets throughout the sample period $[0, T_n]$. Drop out times may also be endogenous – for instance, bankruptcy can trigger a delisting. This concern is discussed in Aït-Sahalia, Jacod, and Xiu (2021) and does not actually create any problems with respect to identification or estimation. It does, however, require additional notation to handle which I omit for the sake of brevity.

3.2 Estimation

To estimate continuous and (semi)jump risk premia, I use the Continuous-Time Fama-MacBeth regression from Aït-Sahalia, Jacod, and Xiu (2021). This procedure is similar to the traditional Fama-MacBeth (1973) estimator in that it consists of time-series regressions followed by cross-sectional regressions. However, in the time-series step, I estimate the more granular continuous and jump betas by running OLS-style rolling regressions on the identified continuous and jump returns, respectively. I then use the estimated betas in monthly cross-sectional regressions to estimate the corresponding premia of the risk factors. These risk premia estimates are averaged over time to obtain an estimate of the true risk premia, Λ_{T_n} . The variance of the risk premia estimate, $\hat{\Lambda}_n$, is readily estimated using the realized sample variance of the spot risk premia estimates. I discuss the procedure more formally below.

¹⁴ Technically, my jump beta estimation approach is not entirely consistent with Aït-Sahalia, Jacod, and Xiu (2021) who assume constant jump betas over their full sample. Instead, my approach is heuristic – I instead estimate jump betas over rolling windows which allows them to vary over time. This procedure allows me to construct a more realistic measure of jump risk.

3.2.1 Time-Series Regressions

To start, I describe the estimation of the jump betas. These betas are estimated using standard multivariate jump regressions (see e.g., Li, Todorov, and Tauchen (2017)). In order to accommodate time-variation, I reestimate the betas on monthly basis over one-year backward-looking rolling windows. Each regression consists of H regressors corresponding to the H jump risk factors. So, for instance, using the Fama-French three factor model with semijumps implies $H = 6$ regressors with each regressor being defined as the up/down jumps of each of the three factors.

More precisely, the (semi)jump betas for some asset m and period i are defined as:

$$\tilde{I}_i = \{j \in \{1, \dots, n\} : i\Delta_n - 1 \leq j\Delta_n \leq i\Delta_n\} \quad (16)$$

$$\hat{R}_{j,k,pos} = \Delta_j^n F^k \cdot \mathbb{1}_{[\Delta_j^n F^k > u_n^f(j,k)]}$$

$$\hat{R}_{j,k,neg} = \Delta_j^n F^k \cdot \mathbb{1}_{[\Delta_j^n F^k < u_n^f(j,k)]}$$

$$\hat{R}'_i = \begin{cases} \left[\sum_{j \in \tilde{I}_i} \hat{R}_j^\top \hat{R}_j \right]^{-1} & \text{if } \zeta \left(\sum_{j \in \tilde{I}_i} \hat{R}_j^\top \hat{R}_j \right) > 1/v_n \\ \mathbf{0}_{H \times H} & \text{otherwise} \end{cases} \quad (17)$$

$$\hat{\beta}_{i,m}^J = \hat{R}'_i \left(\sum_{j \in \tilde{I}_i} \hat{R}_j \Delta_j^n P_m \cdot \mathbb{1}_{[|\Delta_j^n P_m| > u_n^s(j,m)]} \right) \quad (18)$$

where $\hat{\beta}_{i,m}^J$ is a H -dimensional vector. The regression itself is performed over the interval I_i , which encompasses a one-year backward-looking window $(i\Delta_n - 1, i\Delta_n)$. The regressors are the identified semijumps for the set of factors, while the regressand consists of the identified jumps for asset m .¹⁵ The definitions for the case with pure jump betas, rather than semijump betas, are analogous.

The notation above still simplifies some elements of the estimation procedure. Firstly, some stocks have finite lifespans, which requires some additional care. For example, if a stock comes into existence at time t , it is impossible to estimate a jump regression using a backward-looking window at that time. I handle this by ensuring that $\hat{\beta}_{i,m}^J$ is only estimated when there exists a non-trivial amount of data for asset m in each partition of the observations $\tilde{I}_i$. When this does not hold, I simply drop the beta estimate for that asset and period. This approach is not a major concern in practice. Secondly, it is possible for positive jumps to be misclassified as negative jumps if the Brownian motion component is sufficiently large for some interval in finite sample. This is extremely unlikely in practice but is an issue for the asymptotic theory. Handling this issue requires the introduction of additional tuning parameters, which are inconsequential in practice. I leave a discussion of this point in the Online Appendix for brevity.

With the jump betas defined, I now discuss the estimation of the continuous betas. I begin by estimating the spot volatilities and covolatilities using truncated returns over a shrinking window defined by q_n . The spot volatilities of the factors and their spot

¹⁵ Note that there is a tuning parameter v_n that is used to truncate explosive matrix inverses; this is needed for consistency in theory but is inconsequential in a practical implementation. This tuning parameter will also be needed for the continuous beta estimation procedure but will again be inconsequential in practice.

covolatilities with prices are given by

$$\hat{c}_i^{F,k,k'} = \frac{1}{q_n \Delta_n} \sum_{j=0}^{q_n-1} \Delta_{iq_n-j}^n F^k \Delta_{iq_n-j}^n F^{k'} \cdot \mathbb{1}_{[|\Delta_{iq_n-j}^n F^k| \leq u_n^f(iq_n-j,k), |\Delta_{iq_n-j}^n F^{k'}| \leq u_n^f(iq_n-j,k')]} \quad (19)$$

$$\hat{\gamma}_i^{m,k} = \frac{1}{q_n \Delta_n} \sum_{j=0}^{q_n-1} \Delta_{iq_n-j}^n P^m \Delta_{iq_n-j}^n F^k \cdot \mathbb{1}_{[|\Delta_{iq_n-j}^n P^m| \leq u_n^s(iq_n-j,m), |\Delta_{iq_n-j}^n F^k| \leq u_n^f(iq_n-j,k)]} \quad (20)$$

where $i \in \{0, \dots, [n/q_n] - 2\}$. The continuous beta estimate is similar to the usual OLS estimator but using spot (co)volatilities to nonparametrically estimate the true beta.

$$\hat{\beta}_i^C = \begin{cases} \hat{\gamma}_i (\hat{c}_i^F)^{-1} & \text{if } \zeta(\hat{c}_i^F) > 1/v_n \\ \mathbf{0}_{M \times K} & \text{otherwise} \end{cases} \quad (21)$$

In theory, I need the window parameter to be $q_n \asymp \Delta_n^{-\varpi'}$ with $\varpi' > 0$ for consistency, although this does not give any information about its size in a finite sample. So, in practice, I set the parameter such that the window $q_n \Delta_n$ corresponds to 30 calendar days of returns, consistent with suggestions from past work (see Reiß, Todorov, and Tauchen (2015) and Kalnina (2022)). Lastly, as with the jump betas, I need to ensure that this regression is not done on assets with missing data. Hence, I only estimate continuous betas for assets that have data over the full, one-month, backward-looking window; this adjustment is essentially the same as that used with the jump betas.

3.2.2 Cross-Sectional Regressions

In order to estimate risk premia, I run cross-sectional regressions using the estimated betas. To do so, I begin by concatenating the continuous and jump betas to form an estimate for β_i ,

$$\hat{\beta}_{i,m} = \begin{cases} (\hat{\beta}_{i,m}^C, \hat{\beta}_{i,m}^J) & \text{if } \hat{\beta}_{i,m}^C \text{ and } \hat{\beta}_{i,m}^J \text{ exist} \\ \mathbf{0}_{K+H} & \text{otherwise,} \end{cases} \quad (22)$$

where i refers to a particular interval and m refers to a particular asset.¹⁶ I then define $\hat{\beta}_i$ as the $M \times (K + H)$ matrix obtained by stacking each assets' betas. Finally, I use standard cross-sectional regressions to estimate the spot risk premia for each month. All

¹⁶ Note that it is impossible to estimate β_i^C or β_i^J for some assets in particular periods. Whenever this occurs for either set of betas in period i for some asset m , I replace the corresponding row in $\hat{\beta}_{i,m}$ with a vector of zeros; this procedure is equivalent to dropping the associated asset m for period i from the cross-sectional regression defined below.

in all, I have:

$$\hat{\eta}_i = \begin{cases} (\hat{\beta}_i^\top \hat{\beta}_i)^{-1} \hat{\beta}_i^\top & \text{if } \zeta(\hat{\beta}_i^\top \hat{\beta}_i) > 1/v_n \\ \mathbf{0}_{(K+H) \times M} & \text{otherwise} \end{cases} \quad (23)$$

$$\hat{U} = \frac{1}{T_n} \sum_{i=1}^{\lfloor n/q_n \rfloor - 1} \hat{\eta}_i (P_{(i+1)q_n \Delta_n} + P_{iq_n \Delta_n}) \quad (24)$$

$$\hat{U}' = \frac{q_n \Delta_n}{T_n} \sum_{i=1}^{\lfloor n/q_n \rfloor - 1} \hat{\eta}_{n,i} \bar{r}_{iq_n \Delta_n} \quad (25)$$

$$\hat{\Lambda} = \hat{U} - \hat{U}' \quad (26)$$

where the parameter of interest is $\hat{\Lambda}$, a $K + H$ dimensional vector of the first moments of the risk premia for each risk factor.

The variable $\hat{U}$ is a sum of spot risk premia estimated using monthly cross-sectional regressions with estimated betas lagged by one month. Assets that are missing betas are dropped from the regression through the beta zeroing procedure described in [Equation 22](#). This ensures that the cross-sectional regression is well-defined – it is only run on assets where both lagged continuous and lagged jump betas are available. Also, there are no lagged beta estimates at time zero, hence the summations begin at time $q_n \Delta_n$. The variable $\hat{U}'$ is defined similarly but uses regressions on the observed risk-free rate $\bar{r}$ for each interval.¹⁷ Finally, the full risk premia estimate is simply defined as the time-average of the monthly “spot” risk premia estimates.

For inference, we need a central limit theorem for the risk premia estimates $\hat{\Lambda}$. Using the previously stated assumptions along with some other technical assumptions (see [Appendix Section A.2](#)), I have,

$$\begin{aligned} \Delta_{q_n} \bar{P}_i &= (\Delta_n P_{iq_n \Delta_n + \Delta_n}, \dots, \Delta_n P_{(i+1)q_n \Delta_n}) \\ \hat{V} &= \sum_{i=0}^{\lfloor n/q_n \rfloor - 2} \hat{\eta}_i (\Delta_{q_n} \bar{P}_i) (\Delta_{q_n} \bar{P}_i)^\top (\hat{\eta}_i)^\top. \end{aligned} \quad (27)$$

$$T_n \hat{V}^{-1/2} (\hat{\Lambda} - \Lambda_{T_n}) \xrightarrow{d} \mathcal{N}(0, I_{K+H}). \quad (28)$$

This CLT is based on Theorem 1 from Aït-Sahalia, Jacod, and Xiu (2021). Like in the standard Fama-MacBeth estimator, the variance of the risk premia is the (realized) covariance matrix of the spot risk premia estimates. There are some additional technical conditions needed for this CLT to make sense, but I leave them in the [Appendix Section A.2](#) for brevity.

4 Continuous and Jump Risk Premia

The continuous and (semi)jump components of the factor zoo are essentially novel risk factors with presently unknown risk premia. Hence, in this section, I estimate the con-

¹⁷In the regressions, I use the one-month Treasury bill return obtained from Ken French’s website as a proxy for the risk-free rate.

tinuous and semijump risk premia for all the factors from the literature in my dataset. I begin with estimates the workhorse factor models (CAPM, FF3, FF5, and FF6) and their associated continuous and (semi)jump risk premia. I then estimate risk premia for all of the 218 JKP and CZ factors. I finish with risk premia estimates for representative portfolios formed using the factor clusters defined earlier.

4.1 Workhorse Factor Models

Extending the traditional CAPM model by estimating separate continuous and jump betas allows it to more accurately model systematic risk (Todorov & Bollerslev, 2010) and better explain cross-sectional variation in expected returns (Bollerslev, Li, & Todorov, 2016). Motivated by these findings, I similarly extend the workhorse factor models to study how more granular betas affect their performance.¹⁸ To this end, I run Cts-Time Fama-MacBeth regressions on each of the Fama-French workhorse factor models using, as test assets, my set of 272 high-frequency portfolios and the top 100 stocks by market cap each year. I report my results for the continuous/semijump risk factor specification in Table 3. For brevity, I leave the results for the continuous/jump specification along with results for a lower-frequency Fama-MacBeth regression in the Appendix (see Table A.4 and Table A.3, respectively).

Only two risk factors stand out. The first is the negative jump risk factor for the market portfolio. This factor draws a statistically significant, annualized premia of 3.40%, explaining more than a third of the average excess return on market portfolio itself, 7.8%. Moreover, the total jump risk premium on the market, computed as the sum of the semijump risk premia and also estimated directly in Table A.4, is 6% per year. In other words, the vast majority of the premium earned from holding the market comes from jump risk. This finding is broadly consistent with past work – see, e.g., high-frequency studies by Bollerslev and Todorov (2011b), Andersen, Fusari, and Todorov (2016), and Bollerslev, Patton, and Quaedvlieg (2022) along with related work on rare disasters by Rietz (1988), Barro (2006), and Gabaix (2012).

Conversely, the estimated risk premium of the continuous component of the market return is statistically and economically insignificant. This finding is interesting because diffusive moves make up a large portion of the intraday variation in each of the factors. Consequently, one would naturally suspect that their premium would be correspondingly large given the usual intuition from the risk-return trade-off implied by no-arbitrage. However, this conjecture is not supported by the results shown here nor in earlier work done by Bollerslev, Li, and Todorov (2016) and Alexeev, Dungey, and Yao (2017), who have similarly found that continuous market betas, unlike discontinuous market betas, are associated with essentially zero premia.

¹⁸ Aït-Sahalia, Jacod, and Xiu (2021) run similar regressions on their high-frequency Fama-French factors. I redo the regressions here since I need to reference the associated results again in the sequel. Additionally, unlike the regressions from Aït-Sahalia, Jacod, and Xiu (2021), mine include a large set of high-frequency factors as test assets, revealing whether the Fama-French factors can better explain the factor zoo in continuous-time, and also employ rolling jump betas, thereby modeling systematic jump risk with greater flexibility.

The second risk factor of interest is the continuous component of the operating profitability factor, RMW. The magnitude of its risk premia, 4.09%, is about the same as that of the return on the portfolio itself, 3.85%, and its FF3 alpha, 4.68%. Surprisingly, its total jump risk premium, reported in [Table A.4](#), is also statistically significant and non-trivial. In contrast, all of the remaining risk factors draw statistically insignificant risk premia. These null results not particularly surprising given that FF6 portfolios themselves have relatively low average returns within my sample – the average, annual returns for SMB, HML, CMA, and UMD are 1.05%, -0.32%, 1.59%, and 3.91%.

Table 3: CTS-TIME FAMA-MACBETH REGRESSIONS – CONTINUOUS AND SEMIJUMP RISK PREMIA

		<i>Specification</i>							
		CAPM		FF3		FF5		FF6	
FF MKT	Continuous	1.80	(0.48)	0.00	(0.00)	1.66	(0.49)	1.67	(0.50)
	Neg Jump	2.75	(1.39)	3.71**	(2.16)	3.25**	(1.97)	3.40**	(2.08)
	Pos Jump	−0.25	(−0.13)	1.81	(1.08)	2.38	(1.46)	2.64	(1.64)
FF SMB	Continuous			0.18	(0.09)	0.37	(0.20)	0.47	(0.26)
	Neg Jump			1.12	(0.92)	1.12	(0.94)	1.41	(1.22)
	Pos Jump			−0.76	(−0.67)	0.10	(0.09)	0.16	(0.15)
FF HML	Continuous			−0.96	(−0.52)	−1.68	(−0.97)	−1.66	(−0.97)
	Neg Jump			0.32	(0.32)	0.09	(0.09)	0.17	(0.18)
	Pos Jump			−0.73	(−0.81)	0.20	(0.23)	0.22	(0.26)
FF RMW	Continuous					4.15***	(3.53)	4.09***	(3.55)
	Neg Jump					1.24*	(1.70)	1.18	(1.61)
	Pos Jump					0.62	(1.05)	0.39	(0.67)
FF CMA	Continuous					0.42	(0.37)	0.83	(0.74)
	Neg Jump					0.77	(1.12)	0.78	(1.15)
	Pos Jump					−1.36**	(−2.18)	−0.93	(−1.50)
FF UMD	Continuous							1.62	(0.75)
	Neg Jump							−0.65	(−0.54)
	Pos Jump							−1.22	(−1.02)
R^2		32.3%		36.8%		39.0%		40.1%	

Note: I report Cts-Time Fama-MacBeth estimates of the annualized risk premia (%) for each factor along with t-statistics in parentheses. The test assets include every portfolio in the factor zoo along with the top 1000 stocks by market cap in each year. The R^2 values report the time-series average of the R^2 estimates for each cross-sectional regression. The regressions include 3394 test assets and the risk premia are averaged over a time span of 24.9 years. The notation *, **, and *** refers to 90%, 95%, and 99% levels of significance respectively.

Table 4: CTS-TIME FAMA-MACBETH REGRESSIONS - EXPLANATORY POWER

<i>Specification:</i>	CAPM	FF3	FF5	FF6	CAPM	FF3	FF5	FF6
	Stocks				Factor Zoo			
Low-Freq	15.39	18.28	19.59	20.26	24.73	33.77	36.56	37.51
High-Freq	17.65	21.55	23.27	24.22	24.16	37.64	42.11	44.40
Cts/Jmp	19.66	23.94	26.03	27.07	26.92	40.31	44.34	45.93
SemiJump	20.21	24.70	27.14	28.38	28.33	40.90	44.74	46.21

Note: I summarize the R^2 estimates from the Cts-Time Fama-MacBeth regressions in [Table A.4](#) and [Table 3](#). There is one additional model type called “High-Freq” which simply involves running a Cts-Time Fama-MacBeth regression without splitting up the continuous and jump components. The R^2 values report the time-series average of the monthly cross-sectional regression R^2 s. I compute separate R^2 s for stocks and the factor zoo. All reported values are percentages.

In order to better understand how continuous and jump risk factors contribute to cross-sectional explanatory power, I summarize the monthly average cross-sectional R^2 statistics for each of the previously discussed models in [Table 4](#). This table also reports R^2 statistics for Cts-Time Fama-MacBeth regressions done without splitting up the continuous and jump components; these are labelled “High-Freq.” A key finding from [Table 4](#) is that the explanatory power increases as we switch to higher frequency data and more granular risk factors. Going from a low-frequency regression to a high-frequency semijump specification increases the R^2 of CAPM by about 4.9% when using stocks as test assets. About half this increase seems to come from simply using high-frequency data, while the other half comes from separating the continuous and jump components. Similarly, the equivalent increase in R^2 when transitioning from a low-frequency to a semijump FF6 model is 8.1% with equal contributions coming from the increased sampling frequency and the more granular risk factors. Interestingly, the increases in explanatory power when transitioning from the jump to semijump specifications only appear marginal, suggesting that most of the gains are coming from splitting continuous/discontinuous moves rather than signing jumps.

As in traditional regressions, increasing the number of factors also increases the R^2 . Going from CAPM to FF6 improves the R^2 for stocks by about 5% for the low-frequency regression, while the same increase for the semijump regression is about 8%. These increases are more than twice as large when using the portfolios as test assets, likely due to the fact that the factor zoo requires non-market factors to fully span. Lastly, note how the R^2 estimates change along the off-diagonal. The SemiJump CAPM model achieves an R^2 of 20.21% for stocks – this R^2 is about as large as that for the Low-Freq FF6 model, 20.26%. In other words, simply splitting up the market factor gives us about as much explanatory power as adding 5 entirely new factors. On the other hand, the same claim does not hold when using the factor zoo as test assets – the equivalent numbers in this case are 28.33% and 37.51%. This latter finding is likely driven by the fact that non-market factors tend to have very little correlation with the market portfolio, implying that they should benefit comparatively more from adding non-market risk factors.

4.2 The Factor Zoo

I now turn to the risk premia estimates for the factor zoo. Naturally, a statistical analysis of hundreds of factors is subject to the multiple testing problem. To alleviate this, I take two approaches. My first approach is directly analyze all factors at once while correcting for multiple hypothesis testing using the Storey (2002) procedure; this approach deals with concerns over false discovery raised by Harvey, Liu, and Zhu (2016) and Harvey (2017). My second approach is to study the risk premia of the “cluster portfolios” defined earlier in Section 2.4.3. Limiting the analysis to a smaller set of representative portfolios drastically reduces the amount of multiple testing. Moreover, it also produces readily interpretable results since the cluster portfolios proxy for the “themed” risk spanned by each of the clusters.

4.2.1 Individual Analysis

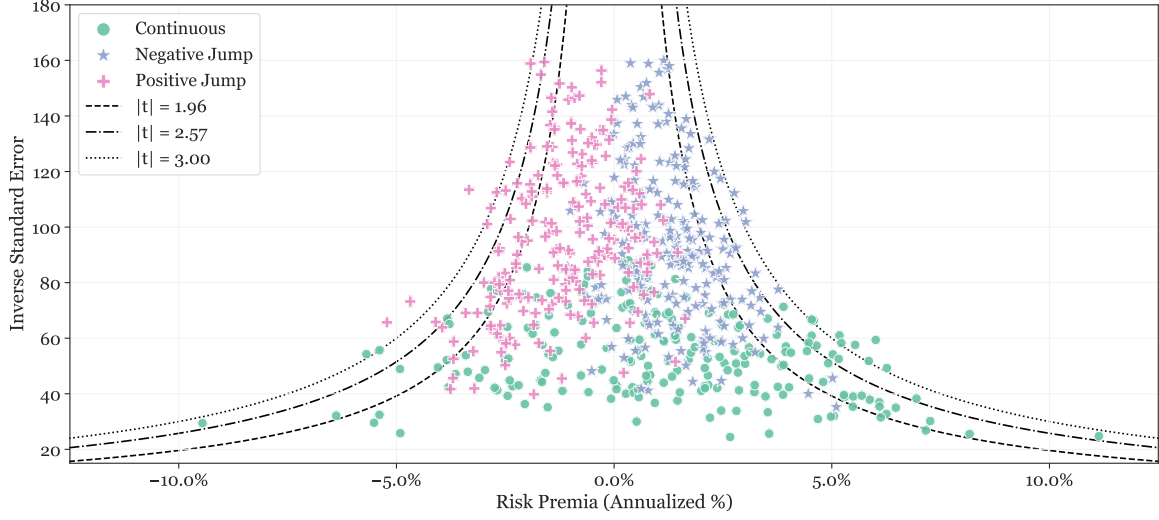
I begin by estimating the continuous and (semi)jump risk premia for each of the 218 JKP and CZ factors. Again, I run Continuous-Time Fama-MacBeth regressions for each factor, reusing the same tuning parameters and test assets from earlier. I also include the FF3 workhorse factors as additional spanning assets in each regression.¹⁹ For brevity, I proceed directly to the continuous/semijump specification where I split each factor into three components and thus use 12 risk factors per regression.

I plot my risk premia estimates in Figure 4. This funnel plot shows the continuous and semijump risk premia estimates for each factor along x-axis paired with their respective inverse standard errors on the y-axis. The plot also marks three funnels for t-statistic cutoffs of 1.96, 2.57, and 3.00. So, for instance, points that lie outside the $|t| = 1.96$ funnel correspond to estimates with t -statistics that exceed 1.96. Overall, there are a total of 133 risk factors that are individually statistically significant at the 5% level and 42 risk factors that are significant at the 1% level. Nineteen estimates lie above the $|t| = 3.00$ threshold mentioned in Harvey, Liu, and Zhu (2016) with ten being continuous risk factors, six being positive semijump risk factors, and three being negative semijump risk factors. Of course, using a threshold of 3.00 is merely a rule-of-thumb adjustment for multiple testing.

In order to more formally handle the multiple testing issue, I compute q -values for each of the estimates (Storey, 2002). These values have a natural interpretation as “posterior Bayesian p-values” (Storey & Tibshirani, 2003) and can be used to perform multiple hypothesis tests while controlling the positive false discovery rate (pFDR) – the expected ratio between the number of false positives to overall positives conditional on rejecting at least one null. More precisely, they represent the smallest pFDR at which an estimate can be called significant. Performing hypotheses tests using q -values is essentially analogous

¹⁹ I explicitly avoid including the remaining FF6 factors – investment, operating profitability, and momentum – to avoid controlling for the same systematic risk that the factors in the zoo aim to capture. For instance, it makes little sense to talk about the premia earned by exposure to a profitability factor when holding exposure to Fama-French RobustMinusWeak constant in a cross-sectional regression. Moreover, the latter three factors of the FF6 were introduced to explicitly capture the “anomalies” raised by the literature – since these are “ex post” factors unlike FF-SMB and FF-HML which have existed for over 30 years, they are particularly inappropriate to control for when studying the zoo itself.

Figure 4: FACTOR ZOO – INDIVIDUAL CONTINUOUS AND SEMIJUMP RISK PREMIA



Note: Each point represents the risk premia estimate for a particular high-frequency risk factor from the set of Jensen, Kelly, and Pedersen (2021) and Chen and Zimmermann (2020) factors along with the inverse standard deviation of the estimate. The estimates are obtained from separate Cts-Time Fama-MacBeth regressions for each factor. The continuous and semijump risk premia estimates are given in differing colors and shapes. The curved lines indicate t-statistic cutoffs. For instance, all points outside the funnel formed by the dashed black line are significant with $p > 0.05$ with respect to a two-sided test of the null. Annotations with factor descriptions are given for estimates with $|t| > 3.00$.

to the Benjamini and Hochberg (1995) method, with the added benefit of reduced Type II error.²⁰

To compute the q-values, I first need to estimate the overall proportion of truly null hypotheses, π_0 . Using the procedure from Storey (2002), which I detail in Appendix Section A.4, I estimate π_0 to be 56%.²¹ Finding that the null is true for 56% (365 out of the 654) of my risk factors is not surprising given the argument from Harvey, Liu, and Zhu (2016) that, “most claimed research findings in financial economics are likely false.” However, it still implies that 289 out of my 654 risk factors have non-trivial risk premia, a seemingly implausibly large number.

In this regard, it is important to recall a point argued by Cochrane (2009) – the statistical or economic significance of a so-called factor’s risk premium does not necessarily imply that it has cross-sectional pricing power. A simple example is an individual stock which may have high returns but naturally lacks the ability to adequately explain the cross-section of expected returns. Another equally important point is that many of the risk factors are quite similar apropos the discussion of the clusters earlier. Consequently,

²⁰ The Benjamini-Hochberg method is implemented by taking a sorted set of p-values, $\{p_1, \dots, p_M\}$, and rejecting the corresponding first m^* hypotheses where $m^* = \max\{m : p_m \leq \gamma m/M\}$. The Storey (2002) procedure is essentially the same but we replace M with M_0 , an estimate of the true number of null hypotheses. By exploiting the fact that not all the hypotheses are actually null, we gain additional statistical power.

²¹ The estimator for π_0 requires a tuning parameter, λ_S . In the Appendix Section A.4, I show that estimates of π_0 are insensitive to this tuning parameter; correspondingly, the q-values are insensitive to the choice of λ_S as well. In addition, using an alternative estimation procedure from Storey and Tibshirani (2003), I find a nearly identical value of $\hat{\pi}_0 = 70\%$, resulting in essentially the same q-values.

Table 5: FACTOR ZOO – SIGNIFICANT CONTINUOUS AND SEMIJUMP RISK PREMIA

Factor Name	Component	Description	Risk Premia	<i>t</i> -stat	q-value
RESFF3.12.1	Pos. Jump	Residual momentum t-12 to t-1	-3.33%	-3.78	0.051
OPE_BE	Continuous	Operating profits-to-book equity	6.01%	3.57	0.051
SEAS.6.10NA	Pos. Jump	Years 6-10 lagged returns, nonannual	-4.68%	-3.43	0.051
INDIPO	Pos. Jump	Initial Public Offerings	-5.22%	-3.43	0.051
EBIT_BEV	Continuous	Return on net operating assets	5.54%	3.19	0.051
FCF_ME	Continuous	Free cash flow-to-price	5.18%	3.16	0.051
FCF_ME	Neg. Jump	Free cash flow-to-price	2.97%	3.14	0.051
AT_TURNOVER	Neg. Jump	Capital turnover	2.76%	3.10	0.051
XFIN	Continuous	Net external financing	-5.69%	-3.09	0.051
EQNETIS.AT	Continuous	Net equity issuance	6.27%	3.09	0.051
MISPRICING_MGMT	Neg. Jump	Mispricing factor: Management	3.03%	3.08	0.051
LTLGR1A	Pos. Jump	Change in long-term investments	-1.92%	-3.05	0.051
NI_BE	Continuous	Return on equity	5.22%	3.04	0.051
OPE_BEL1	Continuous	Operating profits-to-lagged book equity	4.58%	3.04	0.051
FR	Pos. Jump	Pension Funding Status	-2.70%	-3.03	0.051
QMJ	Continuous	Quality minus Junk: Composite	5.85%	3.03	0.051
SALE_BEV	Continuous	Assets turnover	4.54%	3.03	0.051
MEANRANKREVGROWTH	Pos. Jump	Revenue Growth Rank	-2.83%	-3.03	0.051
COMPEQUISS	Continuous	Composite equity issuance	-5.39%	-3.00	0.052

Note: Each row reports the risk premia estimate for a particular high-frequency risk factor from the set of Jensen, Kelly, and Pedersen (2021) and Chen and Zimmermann (2020) factors along with the associated *t*-stat and q-value. The estimates are obtained from separate Cts-Time Fama-MacBeth regressions for each factor. The description columns are based on JKP and CZ. The risk premia estimates are annualized. The associated citations are given in the Online Appendix. The q-values are computed using the procedure given in Storey (2002) which is further detailed in Appendix Section A.4. For brevity, I only report the estimates with *t*-stats greater than 3.00.

the large number of risk factors with non-trivial premia may simply be “rediscoveries” of a sparser set of underlying factors that span their common variation. I will tackle this latter point in the Section 4.2.2 and return to the former point in Section 5.

For now, I proceed with my analysis, using the estimated proportion of nulls, $\hat{\pi}_0$, to compute q-values for each of my estimates. For reference, the q-value for hypothesis $m \in \{1, \dots, M\}$ is calculated as:

$$q(m; \pi_0) = \begin{cases} p(m) \cdot \pi_0 & \text{if } m = M \\ \min(q(m+1), p(m) \cdot \pi_0 \cdot M/m) & \text{otherwise} \end{cases}$$

where $p(m)$ is the m 'th p-value (in ascending order) corresponding to a two-sided test of the m 'th risk premia estimate. I report my results in Table 5 which shows the risk premia estimates sorted by their p-values (and thus their q-values as well). I also limit the table to the estimates that pass a $|t| = 3.00$ threshold for brevity but report the full table in the Online Appendix.

Surprisingly, I find that none of the factors obtain a q-value below 5%. It would then superficially seem that all non-market factor risk is unpriced. However, it is quite unlikely that this is the case given that the estimated proportion of true nulls, $\pi_0 = 56\%$, is both a fairly precise estimate (95% CI of [47%, 65%]) and far from 100%. A better explanation for these statistical findings is that I am simply incurring a large amount of Type II error by using a pFDR threshold of 5%. Therefore, with the aim of making a more reasonable

trade-off between Type I and II error, I instead consider a pFDR threshold of 10%.²² Rejecting all hypotheses with $q < 0.10$, equivalent in this sample to rejecting whenever $|t| > 2.49$, leads me to reject the null for a total of 47 risk factors. The increase in rejections imply a considerable gain in statistical power despite the marginal increase the expected number of false rejections, $47 \times 10\% = 4.7$.

Of these 47 statistically significant risk factors, I find that 12 are negative jump risk factors, 14 are positive jump risk factors, and 21 are continuous risk factors. Moreover, I also find that corresponding risk premia estimates themselves are economically significant. More precisely, the magnitudes of the statistically significant ($q < 10\%$) negative jump, positive jump, and continuous risk factors are 2.84%, -3.00%, and 3.31%, respectively. Relative to the average of FF3 alphas, 1.87% as reported earlier in Table 1, these risk premia estimates are substantial. The fact that semijump risk appears to be priced is also consistent with recent work by Lin and Todorov (2019) and Jacod, Todorov, and Lin (2022) who also argue that systematic non-market jumps and the asymmetry they induce have non-trivial pricing implications.

In addition to magnitudes, the signs of the risk premia estimates generally line up with expectations. All of the statistically significant negative jump risk premia are positive, corroborating the idea that investors receive compensation for exposure to factor “crashes” (see also Chabi-Yo, Huggenberger, and Weigert, 2021). Likewise, all but one of the positive jump risk factors draw a negative risk premium.²³ And, all but four of the continuous risk premia estimates are positive. Since the underlying factor portfolios are long in the same direction proposed by the literature, the simplest explanation for the four incorrectly signed continuous risk premia estimates is that they are likely false positives with their true risk premia being null.

Given that there is a moderately large number of statistically significant continuous risk factors (21 in total), it would seem that continuous factor risk is quite important. However, this finding seems to be primarily driven by “rediscoveries” of the same systematic continuous risk. To corroborate this point, I count the number of significant risk factors associated with each of the three risk components and each of the clusters. I report my counts in Table 6. Note that 7 of the statistically significant ($q < 10\%$) continuous risk factors belong to the Profitability cluster while 4 belong to Low

²² “More reasonable” here is subjective since there does not necessarily exist an optimal trade-off between Type I and II error. The reader is free to choose their own pFDR threshold, and I therefore report all the risk premia estimates sorted by q-value in the Online Appendix. However, it should be noted that standard 1%, 5%, and 10% thresholds used for p-values are not directly translatable to q-value thresholds. This is because the q-value controls the pFDR which is the probability that a risk factor is null *conditional* on its premia being statistically significant; the p-value controls the probability that a premia is statistically significant *conditional* on the risk factor being null. These two probabilities represent vastly different concepts and traditional p-values tell us little about the probabilities of the null and alternative hypotheses being true; see Harvey, Liu, and Zhu (2016) for a discussion of this point. Also, for reference, Barras, Scaillet, and Wermers (2010), who similarly use the Storey procedure but to evaluate mutual funds, consider pFDR thresholds of 5%, 10%, 15%, and 20%.

²³ This need not always be the case in theory. That is, the positive jump risk premium represents compensation for exposure to both systematic jump variation (jumps in general) and good news (positive return). If the negative premia earned for exposure to good news dominates the positive premia earned from exposure to jump variation, then the total positive jump risk premia is negative; otherwise, it can be positive. The question of dominates is ultimately empirical.

Table 6: FACTOR ZOO – NUMBER OF SIGNIFICANT CONTINUOUS AND SEMIJUMP RISK PREMIA BY CLUSTER

Cluster	Total	Significant ($q < 10\%$)			Significant ($q < 15\%$)		
		Cts.	Neg. Jump	Pos. Jump	Cts.	Neg. Jump	Pos. Jump
Value	25	2	1	2	3	3	5
Investment	23	0	1	1	1	6	9
Low Risk	30	4	3	1	17	5	9
Profitability	15	7	2	1	8	6	1
Quality	25	4	2	0	9	4	2
Leverage	29	3	2	1	7	9	6
Momentum	15	0	1	1	4	1	2
Size	12	0	0	0	3	3	3
Profit Growth	13	1	0	1	3	2	1
Accruals	5	0	0	0	0	2	1
Debt Issuance	6	0	0	1	0	1	2
Skewness	6	0	0	3	0	1	6
Seasonality	14	0	0	2	1	3	4

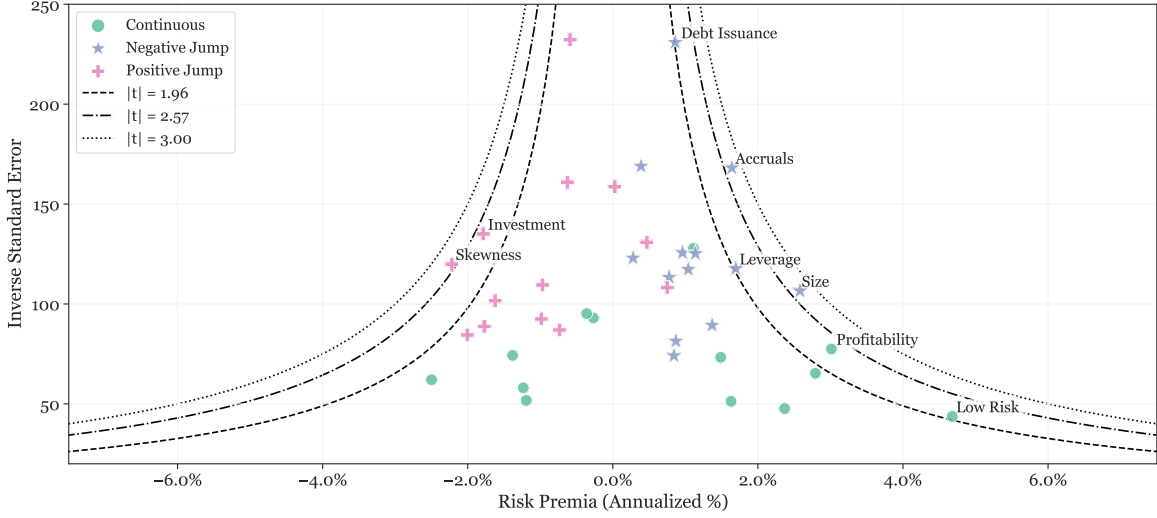
Note: The first data column reports the total number of factor portfolios assigned to each cluster. Each of the remaining columns report the number of statistically significant risk premia associated with each cluster (row) and component (column). The risk premia estimates correspond to the continuous, negative jump, and positive jump components of the high-frequency JKP and CZ factor portfolios. The estimates are obtained from separate Cts-Time Fama-MacBeth regressions for each factor. Statistical significance is determined by two q-value thresholds: 10% and 15%. The q-values are computed using the procedure given in Storey (2002) and further detailed in Appendix [Section A.4](#).

Risk and another 4 belong to Quality. In contrast, the significant positive/negative jump risk factors are fairly evenly distributed across clusters. A similar point holds under a more lenient pFDR threshold of 15% where 17, 8, and 9, and 7 of the statistically significant continuous risk factors are associated with the Low Risk, Profitability, Quality, and Leverage clusters. Likewise, for the negative jumps, 9 of the significant risk premia are associated with the Leverage cluster, while, for the positive jumps, 9 are associated with the Low Risk cluster and another 9 with the Investment cluster. In short, the significant risk premia estimates for each of the components tend to congregate within subsets of the thirteen clusters. Consequently, the number of significant risk premia estimates is not entirely informative about the number of unique sources of priced factor risk associated with each of the three components. To better address this, I now proceed to an analysis of the cluster portfolios constructed earlier.

4.2.2 Cluster Analysis

In order to reduce the severity of the multiple testing problem and avoid retesting the same systematic variation shared by similar factors, I now analyze the zoo using the cluster portfolios in [Section 2.4.3](#). Like earlier, I run Cts-Time Fama-MacBeth regressions using the Fama-French 3 paired with the individual cluster portfolios as spanning assets. I use the same test assets and tuning parameters as before. For brevity, I work directly with a continuous/semijump specification as in the previous section. In total, this amounts to

Figure 5: CLUSTER PORTFOLIOS – CONTINUOUS AND SEMIJUMP RISK PREMIA



Note: Each point represents a risk premia estimate for a particular cluster portfolio; clusters are based on the methodology described in Section 2.4.2 and their representative portfolios are formed using the first principal component of the factor returns within each cluster. The estimates are obtained from separate Cts-Time Fama-MacBeth regressions for each factor. The continuous and semijump risk premia estimates are given in differing colors and shapes. The solid and dashed black lines indicate t-statistic cutoffs. Labels are included for risk factors with $|t| > 1.96$.

thirteen regressions, one for each of the cluster portfolios.

I plot my results in Figure 5. I find that eight risk factors are (individually) statistically significant at a 5% level while three of these are also statistically significant at a 1% level. Of the risk factors that are significant at a 5% level, four capture negative jump risk, two capture positive jump risk, and two capture continuous risk, again corroborating the claim that semijump risk is priced. And, like before, the semijump risk premia estimates appear to be large in magnitude; the averages of the continuous, negative jump, and positive jump risk premia estimates across the thirteen portfolios are 0.78%, 1.11%, and -0.93%. The negative jump risk premia clearly dominate, especially compared to the average return on the cluster portfolios of 1.12%.

Next, to handle multiple testing, I again compute q-values for each of the risk premia estimates. To compute these values, I first estimate the true proportion of nulls π_0 , which I find to be 22% with a 95% CI of [2%, 44%]. I then compute the q-values and find sixteen significant ($q < 0.10$) risk factors with five being negative jump risk factors, another six being positive jump risk factors, and the last five being continuous risk factors. However, given the limited number of hypotheses being tested, the estimated proportion of null hypotheses $\hat{\pi}_0$ is naturally imprecise and the corresponding q-values may be understated.

Consequently, in Table 7, I report q-values based on as moderately larger, and thus more conservative, values for π_0 . Relying on a substantially larger π_0 value of 40% and a pFDR threshold of 10% leads to rejecting the null for just the eight risk factors. These factors are associated the clusters Accruals, Size, Skewness, Investment, Profitability, Low Risk, Leverage, and Debt Issuance. Each of these clusters individually represent 5, 12, 6, 23, 15, 30, 29, and 6 of the original 218 factors from the literature. The statistically significant continuous risk premia results are also consistent with the earlier individual

Table 7: CLUSTER PORTFOLIOS – CONTINUOUS AND SEMIJUMP RISK PREMIA

Cluster Portfolio	Component	Risk Premia	t -stat	q-value		
				$\pi_0 = 30\%$	$\pi_0 = 40\%$	$\pi_0 = 50\%$
Accruals	Neg. Jump	1.64%	2.76	0.030	0.040	0.050
Size	Neg. Jump	2.58%	2.75	0.030	0.040	0.050
Skewness	Pos. Jump	-2.22%	-2.66	0.030	0.040	0.050
Investment	Pos. Jump	-1.79%	-2.41	0.046	0.061	0.076
Profitability	Continuous	3.01%	2.33	0.046	0.061	0.076
Low Risk	Continuous	4.68%	2.05	0.069	0.091	0.114
Leverage	Neg. Jump	1.69%	1.99	0.069	0.091	0.114
Debt Issuance	Neg. Jump	0.86%	1.99	0.069	0.091	0.114
Quality	Continuous	2.79%	1.82	0.089	0.118	0.148
Low Risk	Pos. Jump	-2.00%	-1.69	0.106	0.141	0.176
Leverage	Pos. Jump	-1.62%	-1.65	0.106	0.141	0.176
Seasonality	Pos. Jump	-1.77%	-1.57	0.109	0.145	0.181
Leverage	Continuous	-2.50%	-1.55	0.109	0.145	0.181
Profitability	Neg. Jump	1.14%	1.43	0.121	0.162	0.202
Debt Issuance	Continuous	1.11%	1.42	0.121	0.162	0.202

Note: Each row reports a risk premia estimate for a particular cluster portfolio; clusters are based on the methodology described in [Section 2.4.2](#) and their representative portfolios are formed using the first principal component of the factor returns within each cluster. I report estimates are obtained from running separate Cts-Time Fama-MacBeth regressions for each factor; for these regressions, the span assets are the FF3 plus a given cluster portfolio. The risk-premia estimates are annualized. The q-values are computed using the procedure given in Storey (2002) under various parameterizations for the true proportion of nulls, π_0 . Only the top fifteen estimates by p-value are included for brevity.

analysis, where I found that many of the statistically significant continuous risk factors were associated with the Low Risk and Profitability clusters. And, the finding that several non-market factors draw a non-trivial risk premia is broadly consistent with past work (see, e.g., Aït-Sahalia, Jacod, and Xiu, 2021; Jacod, Todorov, and Lin, 2022; Lin and Todorov, 2019). To ensure that these results are not sensitive to the choice of tuning parameters, I perform additional robustness checks for all of the cluster risk factors in the Online Appendix. I find that only four risk factors survive all of my robustness checks: negative jumps in Accruals, positive jumps in Skewness, positive jumps in Investment, and continuous returns in Profitability.

Since it appears that “discontinuous” factor risk is priced for some of the clusters, it may also be interesting to pursue estimates of the risk premia associated with the overnight (semi)betas of each factor. To this end, I repeat the Cts-Time Fama-MacBeth regressions used thus far but estimate separate betas for the overnight and intraday returns. More precisely, I estimate five betas for each factor portfolio: an intraday continuous beta, two intraday jump semibetas, and two overnight semibetas. This approach essentially follows Bollerslev, Li, and Todorov (2016) but extended to a multifactor setting with semibetas.

I report my results in the Appendix [Table A.5](#). Given the number of risk premia being estimated (5 betas times 13 clusters equals 65 risk premia estimates), I adjust for multiple testing again using q-values. Surprisingly, the estimated proportion of nulls in

this dataset is 100% and none of the risk factors draw a statistically significant premium even at an extremely liberal 40% pFDR threshold. Moreover, only three of the risk factors are individually statistically significant ($|t| > 1.96$) – these are the continuous risk factors for Profitability and Quality, plus the negative jump risk factor for Debt Issuance. In contrast, directly evaluating the magnitudes of the point estimates tells a relatively clearer story. That is, across all of the factors, the average negative intraday jump risk premia is 0.49% compared to 0.31% for the negative overnight risk premia, while the average positive intraday jump risk premia is -0.46% compared to -0.01% for the positive overnight risk premia. In other words, the intraday jump semibetas are associated with larger premia in absolute terms than the overnight semibetas. Thus, from this angle, intraday discontinuous risk appears more economically significant than overnight risk, although the statistical imprecision makes it difficult to draw any definite conclusion overall.

To conclude, there are only a few clusters that have priced risk components. Three of the priced components for the cluster portfolios correspond to semijump risk while just one captures continuous risk; additionally, the majority of semijump risk premia seems to be obtained from exposure to intraday jumps rather than overnight returns.

5 Decomposing the Cross-Section of Returns

The risk premia estimates discussed in the previous section suggest that investors are willing to pay a premium for hedging against certain types of non-market continuous and (semi)jump risk. However, risk premia estimates on their own say only that much – they do not reveal whether the continuous and (semi)jump risk factors are useful for explaining the cross-section of returns. Hence, in this section, I introduce a variance decomposition to separately identify the cross-sectional explanatory power of the jump and continuous components of the factor models studied thus far. To do so, I exploit the fact that cross-sectional variation in expected returns must necessarily come from either the continuous or the jump component of the SDF (see the earlier [Equation 4](#)). Consequently, by estimating the continuous and jump risk premia earned by various test assets, I can readily construct explanatory power bounds that quantify the upper/lower limits of the cross-sectional variation explained each of the two components. As I will show, nearly all variation in the cross-section of expected returns comes from jump risk.

To motivate the decomposition described above, first recall that the factor model used for prices implies a linear structure for the spot excess return (see [Equation 15](#)). Averaging the spot excess return over time gives the standard, average expected return:

$$\begin{aligned}
\frac{1}{T} \int_0^T (\mu_{m,s} - r_s^f) ds &= \frac{1}{T} \int_0^T \beta_{m,s} \lambda_s ds \\
&= \underbrace{\frac{1}{T} \int_0^T \beta_{m,s}^C \lambda_s^C ds}_{\text{Continuous Premia}} + \underbrace{\frac{1}{T} \int_0^T \beta_{m,s}^J \lambda_s^J ds}_{\text{Jump Premia}}, \\
&\equiv \Gamma_m^C + \Gamma_m^J.
\end{aligned} \tag{29}$$

The terms Γ_m^C and Γ_m^J , are defined as the time-averages the continuous and jump risk premia for assets indexed by $m \in \{1, \dots, M\}$. These two terms may be readily interpreted as the expected excess returns coming from covariance with the continuous and discontinuous components of the pricing kernel. Naturally, cross-sectional variation in expected excess returns, the left-hand-side of Equation 29, must come from variation in Γ_m^C and Γ_m^J , making their estimation critical for a variance decomposition. To this end, I propose the following estimators for these two terms:

$$\hat{\lambda}_i = \hat{\eta}_i \left(P_{(i+1)q_n\Delta_n} + P_{iq_n\Delta_n} \right) \quad (30)$$

$$\hat{\Gamma}_{m,n}^C = \frac{1}{T} \sum_{i=1}^{[n/q_n]-1} \left(\hat{\beta}_{m,i}^C \quad \mathbf{0}_H \right) \hat{\lambda}_i \quad (31)$$

$$\hat{\Gamma}_{m,n}^J = \frac{1}{T} \sum_{i=1}^{[n/q_n]-1} \left(\mathbf{0}_K \quad \hat{\beta}_{m,i}^J \right) \hat{\lambda}_i \quad (32)$$

$$\hat{\Gamma}_n = \begin{pmatrix} \hat{\Gamma}_n^C \\ \hat{\Gamma}_n^J \end{pmatrix} \quad (33)$$

$$\hat{E}(R^e) = \frac{1}{T} \left(P_T - P_0 - q_n \Delta_n \cdot \sum_{i=1}^{[n/q_n]-1} \bar{r}_{iq_n\Delta_n} \right). \quad (34)$$

Here, $\hat{\Gamma}_n^C$ and $\hat{\Gamma}_n^J$ are M -dimensional vectors stacking the continuous and jump risk premia estimates for the test assets; $\hat{\Gamma}_n$ simply concatenates these two vectors. The intuition behind these estimators is straightforward – they are just sample analogues of their targets in Equation 29. In fact, these estimators also converge to their targets at a $\sqrt{T}$ rate and are asymptotically normal; a formal proof is given in the Online Appendix.

Next, to measure explanatory power, I define:

$$\begin{aligned} \tilde{R}_J^2 &= 1 - \frac{\sum_{m=1}^M \left(\hat{E}(R_m^e) - \hat{\Gamma}_m^J \right)^2}{\sum_{m=1}^M \left(\hat{E}(R_m^e) \right)^2} \\ \tilde{R}_C^2 &= 1 - \frac{\sum_{m=1}^M \left(\hat{E}(R_m^e) - \hat{\Gamma}_m^C \right)^2}{\sum_{m=1}^M \left(\hat{E}(R_m^e) \right)^2} \\ \tilde{R}^2 &= 1 - \frac{\sum_{m=1}^M \left(\hat{E}(R_m^e) - \hat{\Gamma}_m^C - \hat{\Gamma}_m^J \right)^2}{\sum_{m=1}^M \left(\hat{E}(R_m^e) \right)^2}, \end{aligned} \quad (35)$$

where $\tilde{R}_J^2$ measures the explanatory power of the jump component, $\tilde{R}_C^2$ of the continuous component, and $\tilde{R}^2$ of both components. The $\tilde{R}^2$ measure is readily interpreted as the standard R^2 value associated with a constrained cross-sectional regression between the expected returns for each asset $\hat{E}(R_m^e)$ and the two components $(\hat{\Gamma}_m^C, \hat{\Gamma}_m^J)$.

To define the bounds on the jump and continuous components, I introduce the sets,

$$\mathcal{S}_J^R = \left[\min \left(\tau \left(\tilde{R}_J^2 \right), \tau \left(\tilde{R}^2 - \tilde{R}_C^2 \right) \right), \max \left(\tau \left(\tilde{R}_J^2 \right), \tau \left(\tilde{R}^2 - \tilde{R}_C^2 \right) \right) \right] \quad (36)$$

$$\mathcal{S}_C^R = \left[\min \left(\tau \left(\tilde{R}_C^2 \right), \tau \left(\tilde{R}^2 - \tilde{R}_J^2 \right) \right), \max \left(\tau \left(\tilde{R}_C^2 \right), \tau \left(\tilde{R}^2 - \tilde{R}_J^2 \right) \right) \right], \quad (37)$$

where $\tau(x) \equiv \min(\max(x, 0), 1)$ is used to truncate values outside 0% and 100%. The set $\mathcal{S}_J^R$ gives the bounds on the explanatory power of the jump component, while $\mathcal{S}_C^R$ gives that for the continuous component. The truncation ensures that the bounds do not lie outside $[0,1]$, although this is not much of a concern in practice.²⁴

To better understand these sets, note that all of the $\tilde{R}^2$ measures are equivalent to R^2 s from constrained regressions on the average returns for a given set of test assets. Consequently, the explanatory power of the jump component can either be thought of as either (i) the explanatory power from just using the jump component as the only regressor or (ii) the increase in explanatory power from adding the jump component to a regression that already includes the continuous component. These two definitions arise because the covariance term in a variance decomposition could be attributed to either of the inputs. Equation 36 and Equation 37 simplify formalize this point and explicitly state the bounds.

5.1 Variance Decomposition Results

To estimate these bounds, I begin by running the usual Cts-Time Fama-MacBeth regressions under two different specifications for the span assets. The first specification uses the workhorse FF6 factors and the second specification uses the market factor plus thirteen cluster portfolios. This latter specification, denoted C14, provides an alternative approach to spanning the systematic risk embodied by the zoo. Otherwise, the regressions themselves are run using the same tuning parameters and test assets as those used earlier in Section 4. After estimating the risk premia and expected returns, I compute the R^2 measures for the stocks and portfolios within my set of test assets. I compute the measures for the two groups both together and separately to check for heterogeneity – that is, whether the variance contributions of the continuous and jump risk premia components differ between different types of assets.²⁵

I report my first set of results in Table 8. The first panel reports the R^2 measures for regressions where the span assets were split into two components, continuous and jump; the second panel is similar but involves span assets split into three components, continuous and semijump. To establish a reference point for the variance contributions, I include the $\tilde{R}^2$ measure which reports the cross-sectional explanatory power of the

²⁴ When running a constrained regression, it is possible for the R^2 to decrease when adding a regressor. Hence, bounding the sets $\mathcal{S}$ is necessary for them to lie completely in $[0\%, 100\%]$. However, the main results will not seem to depend on this truncation.

²⁵ Computing the explanatory power measures also involves filtering out assets that spend fewer than 15 years in the sample. This filter ensures that only assets with precise risk premia estimates are included. The results will not substantially change when using a cutoff of 10 years or 20 years. Also note that the filter does not affect portfolios naturally exist forever although, it does cut down the number of individual stocks available for analysis to 780.

Table 8: VARIANCE DECOMPOSITION RESULTS – CTS-TIME FAMA-MACBETH WITH CONTINUOUS/JUMP RISK FACTORS

<i>Panel A. Using Continuous and Jump Risk Factors</i>						
Assets	Specification	$\tilde{R}^2$	Cts Component		Jump Component	
			Lower	Upper	Lower	Upper
All	FF6	48.74	6.00	11.09	37.65	42.74
All	C14	57.62	9.09	11.21	46.41	48.53
Stocks	FF6	48.39	5.58	10.31	38.08	42.82
Stocks	C14	57.04	8.65	11.33	45.71	48.39
Portfolios	FF6	51.20	8.98	16.57	34.63	42.22
Portfolios	C14	61.71	10.33	12.15	49.56	51.38
<i>Panel B. Using Continuous and SemiJump Risk Factors</i>						
Assets	Specification	$\tilde{R}^2$	Cts Component		Jump Component	
			Lower	Upper	Lower	Upper
All	FF6	49.73	5.87	13.57	36.16	43.86
All	C14	58.13	8.14	14.02	44.12	50.00
Stocks	FF6	49.08	5.59	12.77	36.31	43.49
Stocks	C14	57.54	7.81	13.71	43.83	49.72
Portfolios	FF6	54.28	7.81	19.20	35.07	46.47
Portfolios	C14	62.33	10.42	16.22	46.12	51.92

Note: For each specification listed in the second column, I run a Cts-Time Fama-MacBeth regression to estimate betas and spot risk premia for each month. The span assets include the continuous and jump component of each factor in the specification; the test assets are the yearly top 1000 stocks by market cap and all the portfolios in my factor zoo. In the first panel, the regressions involve continuous and jump risk factors, while, in the second panel, the regressions further involve semijump risk factors. For each group of test assets, I then estimate $\tilde{R}_J^2$, $\tilde{R}_C^2$, and $\tilde{R}^2$. The upper and lower bounds for the explanatory power of the jump and continuous components are given by the sets $\mathcal{S}_J^R$ and $\mathcal{S}_C^R$, which are defined in Equation 36 and Equation 37; these sets are also truncated when they exceed 0% or 100%. All values are reported as percentages.

estimate for $\frac{1}{T} \int_0^t \beta_{m,s} \lambda_s ds$. In theory, if a given factor model holds, the $\tilde{R}^2$ values should be exactly 100%. However, in practice, estimation error in the expected returns, betas, and risk premia prevents a perfect fit of the cross-section. Still, $\tilde{R}^2$ estimates appear to be fairly high with the two multifactor models explaining around half the cross-sectional variation in “All” assets. Also, unsurprisingly, the explanatory power is larger for the portfolios than the stocks, likely due to the fact that stocks have a larger idiosyncratic component.

The last four columns of Table 8 report the $(\mathcal{S}_J^R, \mathcal{S}_C^R)$ bounds from Equation 36 and Equation 37. Here, I find a striking result: the jump component of risk premia does far better in explaining the cross-sectional variation in expected returns than the continuous component. In fact, the lower bound on the explanatory power of the jump component is always larger than the upper bound of the explanatory power of the continuous component. And, although the jump component bounds are not always as large as $\tilde{R}^2$, they are quite similar in magnitude. For example, for the FF6 specification in Panel A with

“All” assets, the upper bound on the explanatory power of the continuous component is 11.09% while the lower bound for the explanatory power of the jump component is 37.65%. Relative to the total explanatory power of the factor model, $\tilde{R}^2 = 48.74\%$, the jump component completely dominates the continuous component. Note that this relative comparison is critical to understanding the economic significance of the magnitudes because none of the factor models can fully explain all variation in expected returns. The results shown in Panel B are similar. The magnitudes of the $\tilde{R}^2$ values are similar, consistent with the earlier results from [Section 4.1](#) where I found only marginal increases in cross-sectional explanatory power from splitting the jump risk factors. And, again, the lower bound on the explanatory power of the jump component is substantially larger than the upper bound on that of the continuous component. Lastly, although I have only considered the FF6 and C14 specifications here, the results for CAPM, FF3, and FF5 are essentially the same. In short, differences in exposure to systematic jump risk explains the majority of cross-sectional variation in expected returns.

5.2 Significance of Risk Premia Components

To further stress the importance of jump risk, I now test whether the continuous and jump risk premia of the test assets are statistically significantly different from zero. To do so, I perform statistical significance tests on the Γ^C and Γ^J estimates. These tests are quite similar to those used earlier on the factor risk premia. Essentially, for each test asset, I construct t -statistics for the continuous and jump risk premia, using the realized volatility of the corresponding spot risk premia estimates to compute standard errors. Then, relying on an asymptotic normality result provided in the Online Appendix, I then test for statistical significance in the usual way using two-sided tests.

I report my results in [Table 9](#). It is clear from the table that there are far more cases of statistically significant jump risk premia than continuous. For instance, based on the FF6 specification in Panel A, I find that 48.22% of the stocks in my sample have statistically significant jump risk premia while only 3.81% have statistically significant continuous risk premia. The results are not as distinct for the factor zoo, but the fraction of assets with significant jump risk is generally non-trivial.

Interestingly, the fraction of portfolios with statistically significant continuous risk premia rises from 8.46% to 20.59% when switching from the FF6 specification to the C14 specification, suggesting that the cluster portfolios may be necessary to span the continuous risk embedded in the factor zoo. And, the fraction of statistically significant continuous risk premia for the factors under the C14 specification appear to exceed the equivalent value for the jump risk premia. This is likely related to the fact that there are a large number of factors in the Profitability and Low Risk clusters; the cluster portfolios thereof were shown to draw large continuous risk premia earlier.

Lastly, a non-trivial fraction of stocks draw statistically significantly different continuous and jump risk premia (i.e., the difference $\hat{\Gamma}_m^C - \hat{\Gamma}_m^J$ is statistically significant). The numbers for the factor portfolios are slightly lower but that may be due to the fact that many of the so-called factors do not draw much risk premia in general – see the earlier descriptive statistics from [Section 2.4.1](#).

Table 9: SIGNIFICANCE OF RISK PREMIA COMPONENTS

<i>Panel A. Using Continuous and Jump Risk Factors</i>				
Assets	Specification	Percent Significant ($p < 0.05$)		
		λ_C	λ_J	$\lambda_J - \lambda_C$
All	FF6	3.81	48.22	8.98
All	C14	8.61	45.88	11.69
Stocks	FF6	1.48	63.40	12.01
Stocks	C14	2.59	59.70	14.05
Portfolios	FF6	8.46	18.01	2.94
Portfolios	C14	20.59	18.38	6.99

<i>Panel B. Using Continuous and SemiJump Risk Factors</i>				
Assets	Specification	Percent Significant ($p < 0.05$)		
		λ_C	λ_J	$\lambda_J - \lambda_C$
All	FF6	3.44	39.24	8.24
All	C14	7.01	37.88	10.09
Stocks	FF6	1.48	51.57	11.46
Stocks	C14	2.77	48.98	12.94
Portfolios	FF6	7.35	14.71	1.84
Portfolios	C14	15.44	15.81	4.41

Note: For each model listed in the second column, I run a Cts-Time Fama-MacBeth regression to estimate betas and spot risk premia for each month. The span assets are given by specification column; the first panel involves regressions splitting the span assets into continuous/jump risk factors while the second panel further involves semijump risk factors. The test assets are the yearly top 1000 stocks by market cap and all portfolios in my factor zoo. Then, for each test asset, I compute the components of expected returns linked to systematic continuous and jump risk using Equation 31 and Equation 32; I drop any assets with less than 15 years of available data. Finally, for each group of test assets, I compute the fraction of assets with significant ($|t| > 1.96$) continuous and jump risk premia. These estimates correspond to the columns λ_C and λ_J . The last column reports the fraction of assets for which the difference in these premia is significant. All values are reported as percentages.

In any case, the results from this section and the last suggest that jump risk premia, in contrast to continuous risk premia, plays a large and distinct role in driving the cross-sectional variation in expected returns and driving the individual expected returns themselves. Consequently, distinguishing the risk associated with the continuous and jump components of factors appears critical for comprehensively modelling systematic risk. Along the same lines, the results more generally stress the importance of studying systematic risk and expected returns in a continuous-time setting where the distinction between continuous and jump returns is well-defined.

6 Conclusion

A large body of work has shown that the continuous and jump returns of the market portfolio represent unique sources of risk with distinct pricing implications. This paper

further extends this point to the many factors proposed by the asset pricing literature using a novel dataset of high-frequency factor returns.

In particular, although there are few priced factors overall, four specific risk factors based on my cluster portfolios stand out in terms of both statistical and economic significance. Two of these factors, based on the Investment and Skewness clusters, draw their premia from positive jumps. One, based on the Accruals cluster, draws its premia from negative jumps. And the last, based on the Profitability cluster, draws its premia from the continuous component of its returns. Additionally, for all of the cluster portfolio based risk factors, I find that the magnitudes of the continuous and semijump risk premia are non-trivial and substantially differ between each of the portfolios. The results for the cluster portfolios are also broadly consistent with an individual analysis of all 218 of my characteristic-sorted portfolios.

In addition to the factor zoo, I also reviewed a set of workhorse models based on the Fama-French factors. In this earlier analysis, I found that negative jump risk for the market portfolio commands a statistically significant risk premium of about 3.4% while market jump risk premium itself appears to be about 6.3%. In other words, nearly all the excess return on the market portfolio comes from jump risk while slightly more than half of this jump risk premia comes from negative jumps. In contrast, the operating profitability portfolio (RMW) draws much of its premia from its continuous component, while the remaining FF6 factors do not draw statistically significant premia. I also find that splitting up the workhorse factors into continuous and (semi)jump components appears to increase their cross-sectional pricing ability. Hence, even in standard models, there is value in treating the continuous and jump components of spanning assets as separate risk factors.

Finally, I perform a decomposition of expected returns to tackle the question posed at the start of the paper: why do some assets offer higher return than others? My results show that the majority of cross-sectional variation in expected returns is explained by variation in jump risk premia. Moreover, I find that jump risk premia is often a statistically significant component of individual expected returns while continuous risk premia is less often so. The main takeaway from these results is that jumps are a critical element of systematic risk. Future work would benefit from identifying what causes various factors to jump; doing so would bring us closer to understanding the central question of what drives cross-sectional variation in returns.

Appendix

A.1 Portfolio Construction Methodology

This subsection contains additional details on how I construct my high-frequency portfolios.

FF6 Factors: To start, I reproduce the Fama-French 5+1 factors as faithfully as possible. I begin by constructing the underlying signals from scratch using Compustat data available through WRDS; all signals are constructed exactly as described in the original papers: Fama and French (1993), Fama and French (1996), and Fama and French (2015). Additionally, because my sample of stocks consists of the Fama and French (1993) universe, my data remains faithful to the original papers as well. Then, I construct sorted portfolios for each factor; the portfolio construction methodology is given in Fama and French (2015) as well as on French’s website. As an illustrative example, consider the operating profitability²⁶ factor RMW. For this factor, I begin by constructing six value-weighted portfolios formed on size and operating profitability: Small Robust, Small Neutral, Small Weak, Big Robust, Big Neutral, and Big Weak. The size breakpoint is the median market cap on the NYSE, while the operating profitability breakpoints split the data into 30%/40%/30%. Then, RMW is constructed as

$$RMW = (1/2)(SmallRobust + BigRobust) - (1/2)(SmallWeak + BigWeak)$$

which is the difference between the average return on two high-operating-profitability portfolios and the average return on two low-operating-profitability portfolios. Portfolio assignments are done at the end of June with rebalancing done on a yearly basis, like the original papers. Note that each of the 2×3 portfolios are constructed using high-frequency returns; this is done by simply selecting all the stocks belonging to a particular portfolio and calculating the value-weighted high-frequency returns. Consequently, the RMW factor I produce is high-frequency as well. The remaining characteristic factors of the FF6 are also produced following the original methodology, which is similar to the methodology discussed above for RMW. Finally, the Fama-French Market factor is constructed by taking the value-weighted average of all the stocks in the universe. All in all, the FF6 factors are reproduced faithfully, so the high-frequency factors’ returns will match those of the low-frequency factors.

JKP+CZ Factors: Next, for the 218 JKP and CZ factors, I take a different approach. Specifically, I rebalance portfolios on a monthly basis and I construct portfolios in a simple and uniform way. Like before, factor portfolios are constructed by subtracting the return on a portfolio that is “high” for a signal with the return on a portfolio that is “low” for a signal. But, unlike Fama and French (1993), I construct the high and low portfolios by taking a value-weighted average of stock returns within the top and bottom terciles for a particular signal. This is the same way Jensen, Kelly, and Pedersen (2021) construct their portfolios, except that they winsorize value-weights while I do not. Their winsorization procedure is primarily motivated by their use of international data where a few stocks tend to dominate the market; since I only consider US stocks, this is not as much of a concern for my dataset. Lastly, I use the exact same stock universe as

²⁶ The definition of operating profitability is the same as the original paper. From French’s website, we have: “OP for June of year t is annual revenues minus cost of goods sold, interest expense, and selling, general, and administrative expenses divided by book equity for the last fiscal year end in t-1.”

before: common stock on NYSE/NASDAQ/NYSEMKT. So, overall, the 218 JKP and CZ factors are constructed using a methodology similar to that in Jensen, Kelly, and Pedersen (2021), but with simple value-weights and using the Fama and French (1993) universe of stocks. Furthermore, due to similarities in the portfolio methodologies, my high frequency factors' returns will look similar to those in Jensen, Kelly, and Pedersen (2021).

Industry Portfolios: Finally, for the 48 industry portfolios, I use essentially the same methodology as above for the JKP+CZ factors. Specifically, I compute value-weighted returns on a monthly basis using the same stock universe as before. By using value-weighted returns and the previous stock universe, I remain consistent with Fama and French (1997). By using a monthly rebalancing procedure, I remain consistent with how I construct the JKP and CZ portfolios.²⁷ Additionally, like Fama and French, I use Compustat and CRSP SIC codes, prioritizing Compustat over CRSP; industry classifications are given on French's website. To ensure the industry portfolios are zero-investment, I further subtract the risk-free rate (1-month Treasury Bill return) obtained from French's website.

Finally, note that I use one methodology for constructing the JKP, CZ, and Industry portfolios but another for the FF6 portfolios. My decision here is motivated by the trade-off between keeping the methodology uniform across factors and keeping the methodology consistent with respect to the original work. Since there are many factors in my dataset and I am not simply doing a pure replication of the original papers, I opt for a uniform methodology for the JKP+CZ+Industry portfolios. However, it is critical that the FF6, being the workhorse factors, are constructed properly. So, with respect to the FF6, I opt for consistency with the original work by Fama and French (2015). I set aside a more extensive argument for this approach to the Online Appendix for brevity. Additionally, in the Online Appendix, I also discuss the accuracy of my JKP and CZ portfolios with respect to their original low-frequency counterparts.

A.2 Assumptions

The Cts-Time Fama-MacBeth regression requires multiple assumptions for identification and estimation. These assumptions come from Aït-Sahalia, Jacod, and Xiu (2021). I begin by listing and discussing the assumptions with straightforward economic content. Then, I list the more technical assumptions.

Primary Assumptions

A.I. Independence of Idiosyncratic Risk: The Brownian motion and Poisson random measure for the idiosyncratic risk component P_t^I are independent of those for the factors: $(W^I, \underline{p}^I)$ is independent with respect to $(W^F, \underline{p}^F)$.

A.II. Factor Structure: Let $\lambda_t^{C,k}$ be the risk premium process for the risk factor $\tilde{F}^k$, $\lambda_t^{J,h}$ be the risk premium process for risk factor $\bar{F}^h$, and $\lambda_t^{I,m}$ be the risk premium process for the idiosyncratic risk. Define λ_t^C and λ_t^J by stacking the risk premium processes for each component; define λ_t similarly. The drift of the

²⁷ Using a monthly rebalancing procedure is an insignificant decision, since firms are unlikely to switch between industry classifications very often. Yearly rebalanced industry portfolios are nearly identical.

log-price processes is given by

$$\mu_t - \bar{r}_t = \beta_t \lambda_t + \lambda_t^I \quad (\text{A.1})$$

for all t .

A.III. Weakly Unpriced Idiosyncratic Risk: Define $\Lambda_T^I = \frac{1}{T} \int_0^T \lambda_t^I dt$. For all $m \in \{1, \dots, M\}$, we have $\sqrt{T} \Lambda_T^{I,m} \xrightarrow{P} 0$ as $T \rightarrow \infty$.

Assumption A.I is similar to an exogeneity condition between the regressors (factors) and residual (idiosyncratic risk). This assumption stronger than the usual orthogonality condition. Still, independence is an expected requirement, since the continuous betas will be estimated nonparametrically using a shrinking window. Additionally, the orthogonality implied by independence is also natural; if the idiosyncratic term was correlated with the factors, it would represent systematic risk.

Assumption A.II is similar to the result from Ross (1976) but far weaker. In actuality, we may obtain the original APT result by making a lower level no-arbitrage restriction. That is, suppose we can define a set of portfolio weights, a M -dimensional predictable process ϕ_t , such that $\phi_t \beta_t = 0$ for all t . A portfolio with these weights would have no exposure to the factors and consequently no systematic risk; a no-arbitrage condition would then force the excess return to be zero. This is essentially the argument from Ross (1976) adapted to the continuous-time setting by Ait-Sahalia, Jacod, and Xiu (2021). However, note that Equation A.1 contains a risk premia term for the idiosyncratic risk component. Within the context of this paper, it is actually not necessary for the idiosyncratic risk to be unpriced in every period.

Instead, Assumption A.III is sufficient for identification; this assumption essentially states that the time-average of the idiosyncratic risk scaled by $\sqrt{T}$ converges to zero. This weaker condition arises because the estimation procedure only involves estimating a time average of risk premia and not necessarily estimating the “spot” risk premia, which is generally not possible without stronger assumptions. Consequently, it is sufficient to have the time average of the idiosyncratic risk premia disappear over a long horizon rather than be set to zero in every period.²⁸ Economically, this condition means arbitrage opportunities can arise in the short-run but cannot be sustained in the long-run. This accounts for possible limits to arbitrage, although the assumption could also be trivially strengthened to a full no-arbitrage condition.

Lastly, it should also be noted that there is no ergodicity assumption here. Such an assumption is optional, because the continuous-time Fama-MacBeth procedure is able to estimate the moving target $\frac{1}{T_n} \int_0^{T_n} \lambda_t dt \equiv \Lambda_{T_n}$. If we invoke ergodicity, we can assume that Λ_{T_n} converges to Λ_∞ and treat $\hat{\Lambda}_{T_n}$ as its estimator. But, this is not necessary; consistent (and asymptotically normal) estimates of the historical risk premia Λ_{T_n} may

²⁸ The intuition behind this assumption comes from the central limit theorem for the risk premia estimator. First, note that I aim to estimate the risk premia of the factors

$$\Lambda_T \equiv \frac{1}{T} \int_0^T \lambda_t dt \equiv \frac{1}{T} \int_0^T (\lambda_t^C, \lambda_t^J) dt. \quad (\text{A.2})$$

Next, note that AJX prove that $\sqrt{T_n}(\hat{\Lambda}_n - \Lambda_{T_n}) = O_p(1)$ where $\hat{\Lambda}_n$ is the estimator (discussed in the next section) and Λ_{T_n} is the target. From here, it is natural to see why $\sqrt{T} \Lambda_T^{I,m} = o_p(1)$. If the time average of the idiosyncratic risk Λ_T^I were any larger, it would contaminate the limiting distribution or break consistency.

also be interesting on their own. Later on, when I define the estimators, I will continue to treat Λ_{T_n} as the estimation target although it could easily be replaced with Λ_∞ . This concludes the primary assumptions; I now list some additional, more technical assumptions.

Secondary Assumptions

Property 1: Let Y_t be some multidimensional, optional process and $\mathcal{L} = (R, R'] \cap (0, \infty)$ be a random interval with $R < R'$ being two stopping times; there is a constant C such that $\forall s \geq 0$ and all finite stopping times S with $R < S \leq R'$, we have:

$$\left\| \mathbb{E}(Y_{(S+s) \wedge R'} - Y_S | \mathcal{F}_S) \right\| \leq Cs, \quad \mathbb{E} \left(\left\| Y_{(S+s) \wedge R'} - Y_S \right\|^2 | \mathcal{F}_S \right) \leq Cs \quad (\text{A.3})$$

This is a higher level property from Aït-Sahalia, Jacod, and Xiu (2021). It will be imposed on the beta and risk-free rate processes.

B.I. Factor Process: The processes μ_t^F and σ_t^F are optional and bounded, $c_t^F = \sigma_t^F (\sigma_t^F)^\top$ is invertible with a bounded inverse, and the function δ^F on $\Omega \times \mathbb{R}_+ \times E$ is predictable and there are a Borel bounded function $\tilde{\Gamma}$ on E and a number $\alpha \in [0, 1)$ such that $\|\delta^F(\omega, t, z)\| \leq \tilde{\Gamma}(z)$ and $\int_E \tilde{\Gamma}(z)^\alpha \nu(dz) < \infty$. Here, E is an arbitrary Polish space.

B.II. Idiosyncratic Risk Process: The processes μ_t^I and σ_t^I are optional and bounded, and we have $\|\delta^I(\omega, t, z)\| \leq \tilde{\Gamma}(z)$ where $\tilde{\Gamma}(z)$ is the same as in the previous assumption.

B.III. Risk-Free Asset Process: The process r_t is optional, bounded, and satisfies Equation A.3 on $\mathbb{R}_+$.

B.IV. Factor Loadings: The process β_t^C is optional and bounded, and the $\mathcal{M}_{K+H}^+$ -valued process $b_t = \beta_t^\top \beta_t$ is invertible with a bounded inverse.

B.V. Jump Partitions: For arbitrary reals $\chi, \chi' > 0$ and $\partial \bar{B}^h$ as the boundary of $\bar{B}^h$, define

$$\bar{B}^h(\chi, \chi') = \{x \in \bar{B}^h : \chi \leq |x| \leq \chi', d(x, \partial \bar{B}^h) \leq \chi'\} \quad (\text{A.4})$$

Then, for each $m \in \{1, \dots, M\}$, $t > 0$, and $\rho' > 4\rho > 0$, define

$$\begin{aligned} A(\rho, \rho')_t^m &= (\zeta_m, \theta_m) \cap \left(\bigcup_{i \leq h \leq H} \{s \in (0, t] : \Delta \bar{F}_s^h \in \bar{B}^h(2\rho, \rho'/2)\} \right) \\ N(\rho, \rho')_t^m &= \#(A(\rho, \rho')_t^m) \end{aligned} \quad (\text{A.5})$$

$R(m, \rho, \rho', t)$ is the $N(\rho, \rho')_t^m \times H$ -matrix with entries

$$R(m, \rho, \rho', t)^{s,h} = \Delta \bar{F}_s^h \text{ for } s \in A(\rho, \rho')_t^m, 1 \leq h \leq H$$

When $H \geq 1$, we have $\Pr(\zeta(R(m, \rho, \rho', t)^\top R(m, \rho, \rho', t)) \geq \varepsilon) \rightarrow 1$ as $t \rightarrow \infty$, for each m and some $\varepsilon, \rho, \rho' > 0$; this also implies $\Pr(N(\rho, \rho')_t^m \geq H) \rightarrow 1$ as $t \rightarrow \infty$.

B.VI. Relative Asymptotic Behavior of Sequences: We have

$$u_n \asymp \Delta_n^\varpi, \quad q_n \asymp \Delta_n^{-\varpi'}, \quad v_n \asymp \log(1/(T_n \Delta_n)) \quad (\text{A.6})$$

for the BV truncation parameter u_n (in the main text, I directly use Δ_n^ϖ), the spot covariance window parameter q_n , and the inverse truncation parameter v_n . Additionally, the sequences above along with Δ_n and T_n satisfy the following conditions:

$$\sup_n \left(T_n \Delta_n^\tau + \frac{1}{T_n \Delta_n^\tau} \right) < \infty, \quad \text{where} \quad \begin{cases} 0 < \tau < 1 & \text{if } H = 0 \\ 0 < \tau < \frac{2}{11} & \text{if } H \geq 1 \end{cases} \quad (\text{A.7})$$

$$\begin{cases} \max\left(\frac{t}{2}, 1 - \tau\right) < \varpi' < 1 - \frac{\tau}{2}, & \text{if } H = 0. \\ \max(5\tau, 1 - \tau) < \varpi' < 1 - \frac{\tau}{2}, & \text{if } H \geq 1. \end{cases} \quad (\text{A.8})$$

$$\begin{cases} 0 < \frac{1}{2} - \varpi < \frac{1-\alpha}{64-2\alpha}, & \text{if } H = 0. \\ 0 < \frac{1}{2} - \varpi < \min\left(\frac{1-\alpha}{64-2\alpha}, (\varpi' - 5\tau), \frac{1-\varpi'}{6}\right), & \text{if } H \geq 1. \end{cases} \quad (\text{A.9})$$

where α comes from Assumption B.I.

B.VII. Conditions on the Variance Sequence: Define

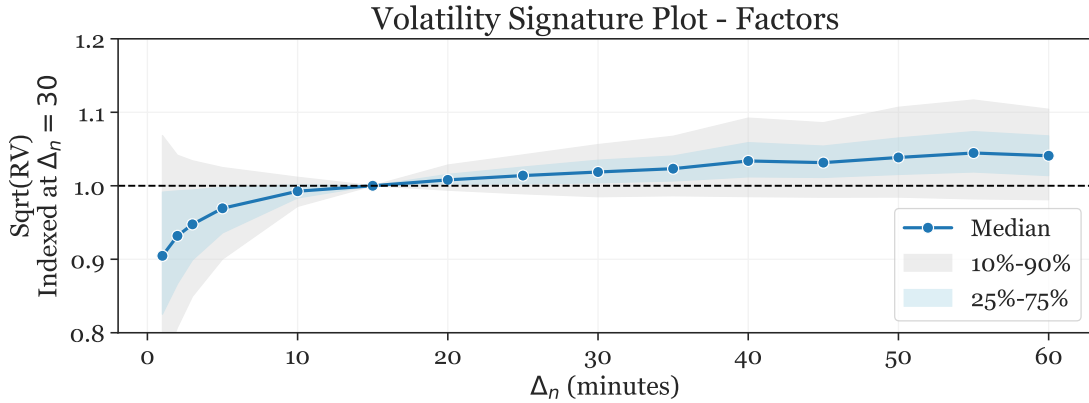
$$V_t^{j,j'} = \int_{m,m'=1}^M \int_0^t \eta_s^{j,m} \eta_s^{j',m'} d[P^m, P^{m'}]_s \quad (\text{A.10})$$

for $j \in \{1, \dots, K + H\}$. Assume that $(1/t)V_t$ converges in probability to a limit V_∞ as $t \rightarrow \infty$.

A.3 Signature Plots

In this section, I provide signature plots for my high-frequency, characteristic-sorted factor portfolios. The factors considered include the six Fama-French factors along with the 218 JKP and CZ factors. Because this set of assets is large, I plot summary statistics for the volatility signatures instead of all 224 signatures. To construct these statistics, I first compute the individual signatures and normalize each by their value at $\Delta_n = 15min$. I then compute the median of the volatility signatures at each Δ_n along with 25-75 and 10-90 quantiles to help visualize any heterogeneity. I plot the indexed signatures in Figure A.1.

Figure A.1: INDEXED VOLATILITY SIGNATURE PLOTS – CHARACTERISTIC-SORTED FACTORS

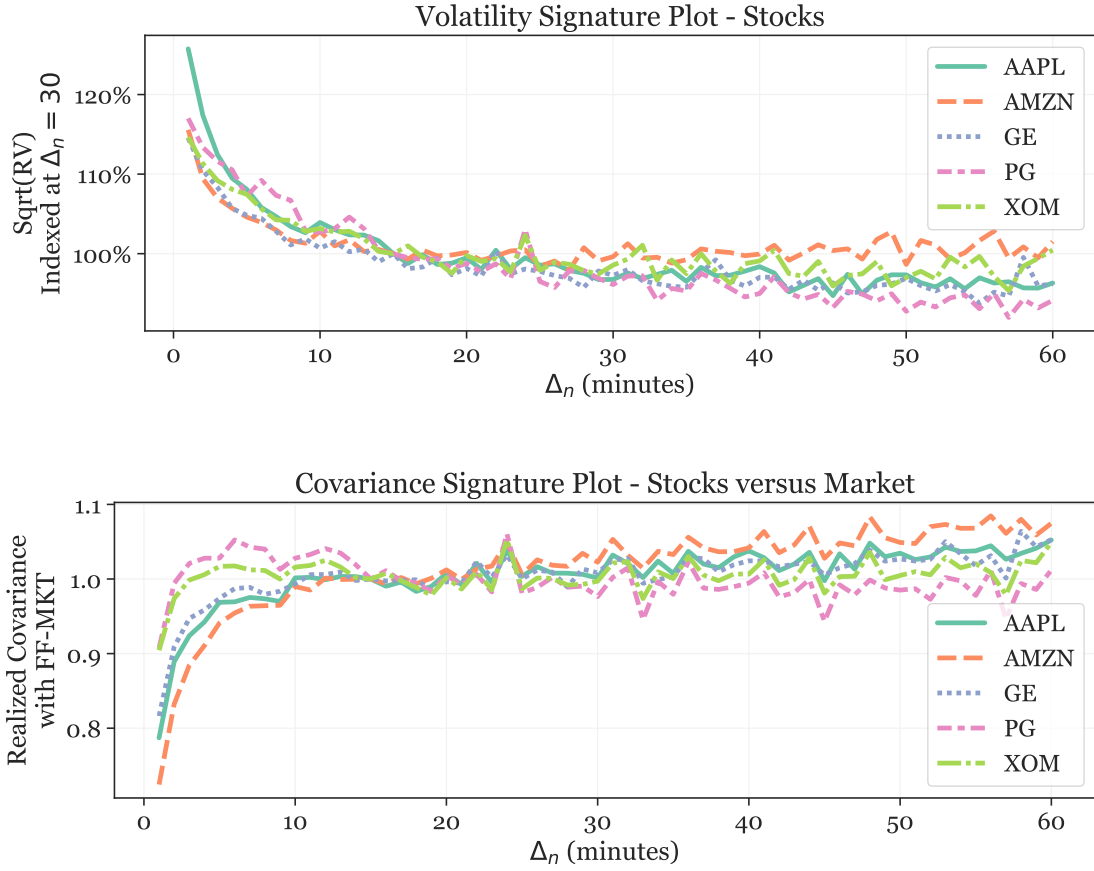


Note: I compute volatility signatures for each of the FF, JKP, and CZ portfolios; I index each signature to 1 at $\Delta_n = 15min$. Then, at each sampling frequency, I calculate the median indexed realized volatility across factors. I also report the 5%-95% and 25%-75% quantiles for this statistic over the cross-section.

Surprisingly, the median signature curves downwards rather than upwards as the sampling frequency approaches zero. Furthermore, based on the plotted quantiles, this behavior is pervasive across the factors. The simplest explanation for why this occurs is the presence of cross-sectionally orthogonal microstructure noise in the stocks underlying the portfolios. At higher frequencies, aggregating the noisy-heavy stock returns into portfolios induces a “diversification” effect that kills the volatility. Some specific sources of noise that can produce this effect are bid-ask bounce, tick size limits, and asynchronous trading. Indeed, all of these sources contribute to the “Epps Effect,” defined as the breakdown of return correlations at high-frequencies (Epps, 1979). Moreover, these signatures are not unique to my factors – the signatures of the high-frequency factors produced in Aït-Sahalia, Kalnina, and Xiu (2020) appear similar.

To further study the level of microstructure noise at various frequencies, I plot volatility and covariance signatures for a few large-cap stocks in Figure A.2. In the first subplot, we can see the usual, upward-curving volatility signature for all five stocks. In the second subplot, which reports the indexed realized covariances with respect to my high-frequency market factor, we can see a downward-curving signature where covariances break down at very high frequencies. These two signatures further corroborate my earlier argument regarding the downward-sloping factor volatility signatures. The underlying return data for stocks exhibit the usual signature, while the downward-sloping covariances are consistent with the downward-sloping factor signatures.

Figure A.2: VOLATILITY AND COVARIANCE SIGNATURE PLOTS – STOCKS



Note: In the first subplot, I compute volatility signatures for Amazon, Apple, General Electric, Exxon Mobile, and Proctor & Gamble. I index each signature to 1 at $\Delta_n = 15min$. In the second subplot, I compute covariance signatures for the same stocks against my high-frequency Fama-French market factor. I index these signatures in the same way. For both plots, I set the sample period to 2000-2020 to ensure a balanced panel.

Besides for the sharp slopes as the sampling frequency rises, all three signature plots exhibit the usual flattening out at lower frequencies. In particular, they flatten out at around 15-minutes. This finding motivates my use of a coarse, 15-minute sampling frequency in the main text to alleviate concerns regarding microstructure noise.

A.4 Multiple Testing Adjustment Details

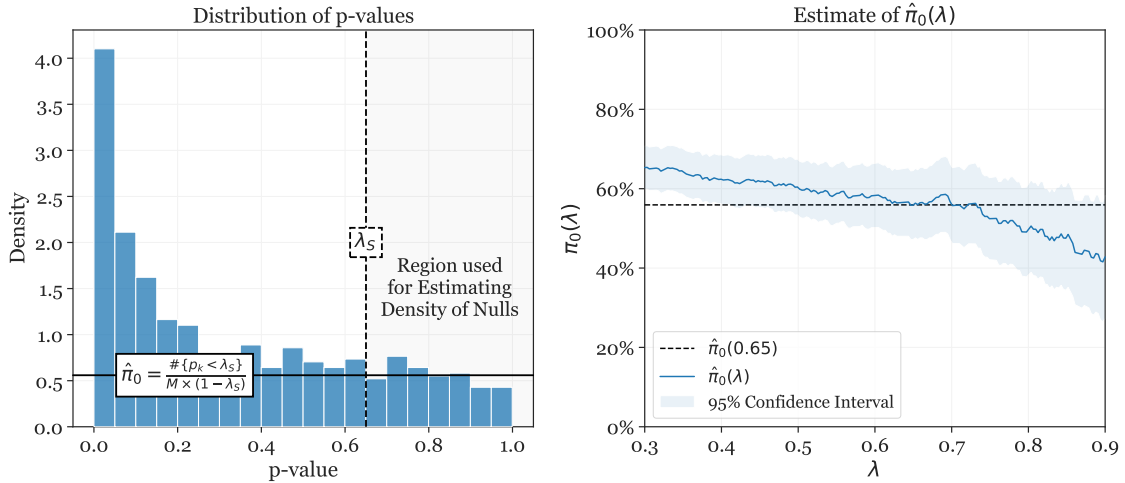
This subsection provides additional details regarding the multiple testing procedure used in [Section 4.2.1](#). This procedure, based on Storey (2002), aims to improve the power of the Benjamini and Hochberg (1995) procedure by estimating the proportion of true nulls and adjusting the BH procedure accordingly.

Naturally, the first step is to estimate the true proportion of nulls: π_0 . To do so, I use Algorithm 1 from Storey (2002) which suggests the following estimator:

$$\hat{\pi}_0 = \frac{\#\{p_k > \lambda_S\}}{M \times (1 - \lambda_S)}. \quad (\text{A.11})$$

Here, M is the number of hypotheses, p_k is the p-value for the k 'th hypothesis, and λ_S is a tuning parameter. The intuition behind this estimator comes from the fact that, for any set of hypotheses, p-values under the null are distributed uniformly while p-values under the alternative cluster around zero. We can exploit this to estimate the proportion of truly null risk premia by estimating the density of the p-value distribution above some sufficiently large threshold λ_S . In order to select λ_S , I follow the bootstrap procedure from Storey, Taylor, and Siegmund (2004) which returns $\hat{\lambda}_S = 0.65$.²⁹

Figure A.3: ESTIMATING THE PROPORTION OF TRUE NULLS



Note: In the first subplot, I produce a histogram showing the distribution of the p-values from the risk premia estimates produced in [Section 4.2.1](#). Each p-value corresponds to a particular factor and either its continuous, positive jump, or negative jump risk premia. The vertical line indicates my primary choice of λ_S for the Storey procedure while the horizontal line is the associated density estimate, $\hat{\pi}_0(\lambda_S)\%$. The second subplot shows how the density estimate varies with the choice of λ_S . The confidence interval is computed using the asymptotic normality result for Storey's estimator given by Proposition 3.2 of Genovese and Wasserman (2004).

To further clarify the procedure and whether the results are sensitive to the choice of λ_S , I plot the density of the p-values for risk premia estimates for the factor zoo from [Section 4.2.1](#) in the first subplot of [Figure A.3](#).³⁰ The dashed vertical line marks my choice $\lambda_S = 0.65$ while the solid horizontal line reports the implied density of null hypotheses,

²⁹ My implementation of the bootstrap procedure follows that of Barras, Scaillet, and Wermers (2010) exactly.

³⁰ The left-hand-side subplot is based on Figure 2 in Barras, Scaillet, and Wermers (2010) who use the Storey procedure to distinguish between lucky, skilled, and unskilled mutual funds.

Table A.1: NUMBER OF SIGNIFICANT RISK PREMIA VERSUS NULL PROPORTION ESTIMATE

Factor Class	π_0					
	0.56	0.60	0.70	0.80	0.90	1.00
Continuous	21	20	16	10	10	10
Neg. Jump	12	11	8	6	4	3
Pos. Jump	14	13	10	8	8	8

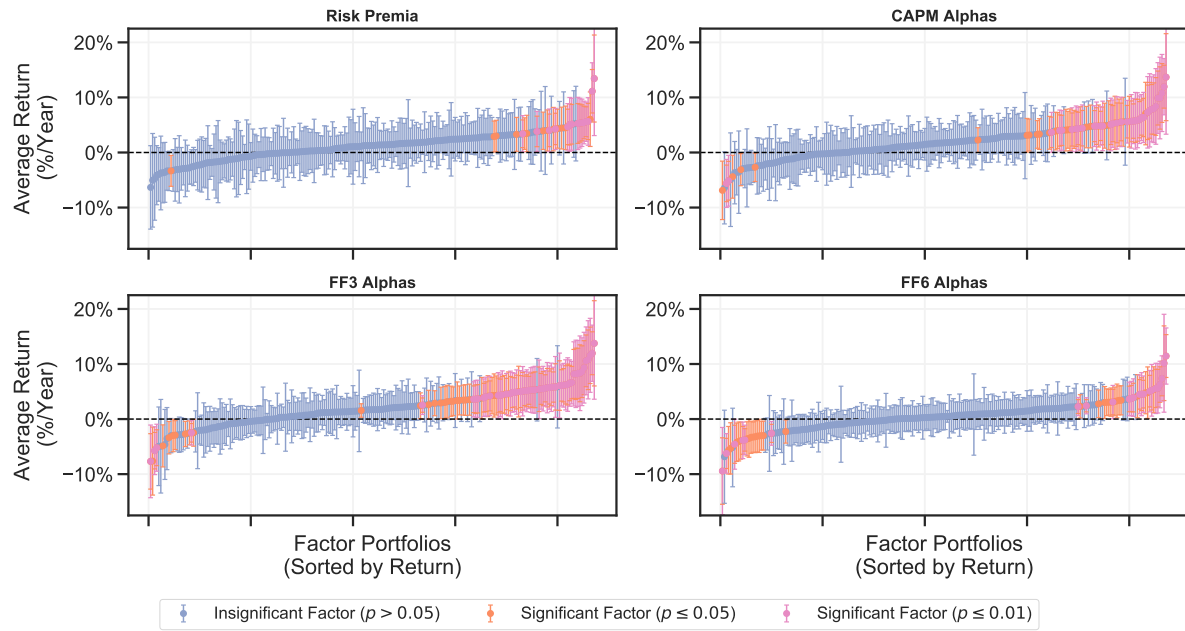
Note: Each entry reports the number of statistically significant ($q < 10\%$) risk premia estimates under a particular value of π_0 associated with a certain class of risk factors. The risk premia estimates themselves are obtained by running Cts-Time Fama-MacBeth regressions on each of the 218 JKP and CZ factor portfolios. Statistical significance is based on q -values which are computed using the procedure given in Storey (2002) and depend on the choice of π_0 .

$\hat{\pi}_0 = 56\%$. The data used for estimated the density simply consists of the p -values larger than λ_S . The second subplot shows that my main estimate, 56%, is quite robust to the choice of λ_S . Moreover, the 95% confidence interval is reasonably tight, suggesting that the estimate of π_0 is fairly precise.

In order to ensure that the number of rejections is not particularly sensitive to the choice of π_0 , I report the number of statistically significant ($q < 10\%$) risk factors across a range of values for π_0 in Table A.1. It is clear from the table that the overall number of rejections is fairly insensitive to the choice of π_0 . Moreover, even when $\pi_0 = 100\%$ and the Storey procedure becomes equivalent the Benjamini-Hochberg procedure, we can still reject 21 risk factors. Additionally, the composition of rejections across the three factor components is similarly insensitive. So, overall, the results do not substantially change even under moderately larger values of π_0 .

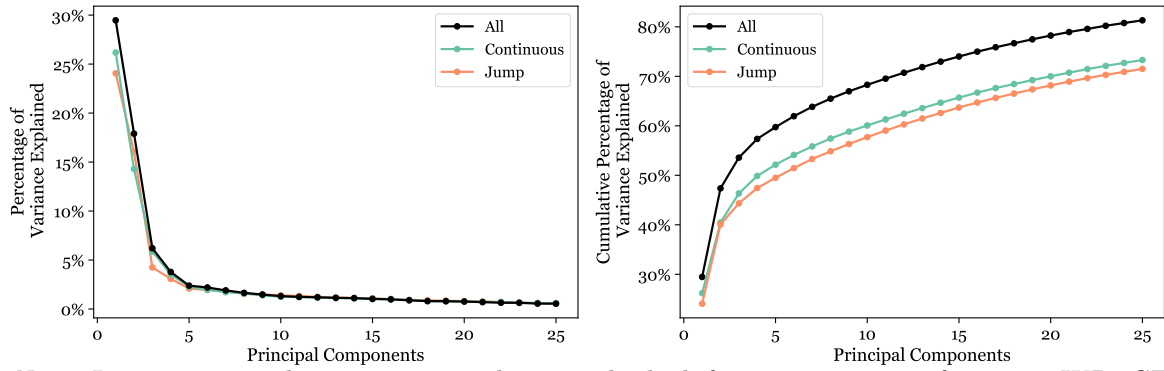
A.5 Additional Tables and Figures

Figure A.4: FACTOR ZOO – RISK PREMIA AND ALPHAS



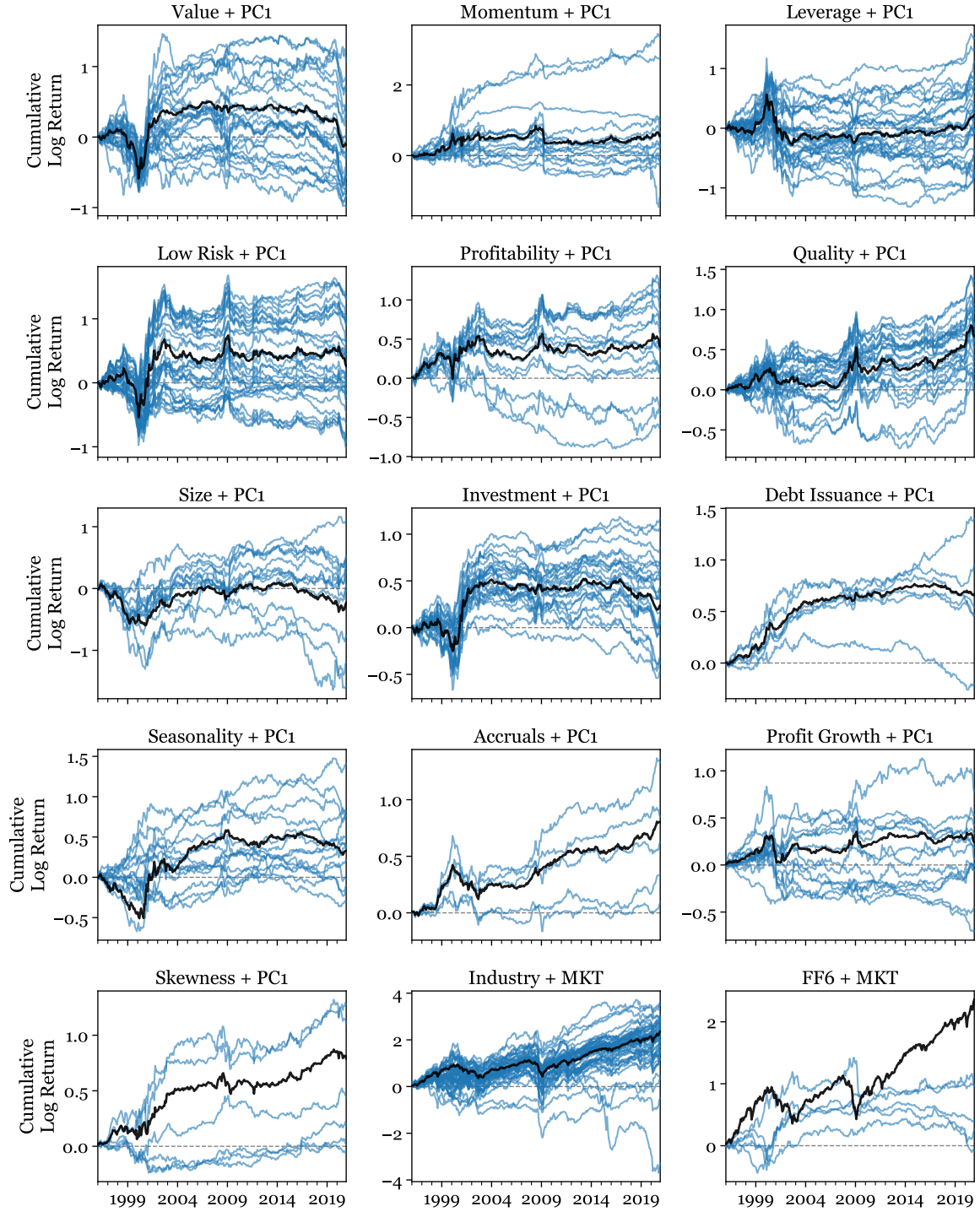
Note: Each subplot shows the returns or alphas for each of the 218 JKP+CZ factors. The industry portfolios and the FF6 are discluded from the figure. Additionally, the factors are sorted by their risk premia or alphas based on the subplot; the signs of the portfolio returns are based on the signs proposed by the original papers. The bars represent 95% confidence intervals. The point estimates and CIs are colored according to statistical significance. The figure style is based on Jensen, Kelly, and Pedersen (2021).

**Figure A.5: FACTOR ZOO – PRINCIPAL COMPONENTS ANALYSIS –
CONTINUOUS/JUMP**



Note: I run a principal components analysis on the high-frequency returns of my 218 JKP+CZ portfolios. The underlying data spans from 1996 to 2020. I also repeat this analysis on the continuous and jump returns with each identified using the standard bipower truncation procedure described in [Section 2.4.3](#). The first subplot reports the portion of total variation explained by each principal component up to 25 in total. The second subplot is similar but reports the cumulative total variation explained.

Figure A.6: FACTOR ZOO AND CLUSTER PORTFOLIO RETURNS



Note: For each of the thirteen clusters defined in [Section 2.4.2](#), I plot the cumulative returns on each of the underlying factors along with their first principal component. The factor returns are in blue while the principal component portfolio (or “cluster portfolio”) is given in black. For clarity, the cluster portfolios have been resigned to covary positively with their underlying factors. The last two subplots also report the returns on my 48 industry portfolios and Fama-French 6 portfolios; in both, the market return is given in black. All returns are monthly and are accumulated over the full sample, 1996-2020.

Table A.2: CLUSTER PORTFOLIOS – COJUMP TESTS

	AAPL	BAC	DIS	GE	GIS	IBM	JNJ	JPM	KO	MMM	MRK	MSFT	PG	WMT	XOM	AMZN
Market	0.84	0.90	0.83	0.90	0.84	0.92	0.91	0.87	0.89	0.90	0.84	0.91	0.89	0.88	0.89	0.83
Value	0.89	0.87	0.88	0.88	0.86	0.90	0.92	0.87	0.88	0.91	0.87	0.94	0.89	0.92	0.87	0.95
Investment	0.91	0.88	0.85	0.87	0.84	0.88	0.87	0.86	0.86	0.87	0.85	0.92	0.91	0.92	0.88	0.94
Low Risk	0.93	0.91	0.86	0.87	0.85	0.92	0.92	0.91	0.88	0.89	0.86	0.92	0.90	0.89	0.86	0.97
Profitability	0.88	0.94	0.92	0.90	0.90	0.93	0.95	0.92	0.91	0.91	0.92	0.92	0.95	0.91	0.93	0.92
Quality	0.85	0.93	0.89	0.93	0.87	0.93	0.93	0.93	0.90	0.89	0.89	0.94	0.90	0.90	0.89	0.94
Leverage	0.92	0.89	0.88	0.89	0.87	0.91	0.91	0.87	0.88	0.90	0.86	0.94	0.90	0.89	0.88	0.97
Momentum	0.90	0.92	0.90	0.94	0.87	0.93	0.91	0.92	0.92	0.89	0.88	0.92	0.89	0.94	0.88	0.94
Size	0.86	0.90	0.85	0.90	0.88	0.90	0.92	0.87	0.90	0.89	0.87	0.93	0.92	0.91	0.88	0.90
Profit Growth	0.86	0.91	0.87	0.89	0.87	0.90	0.90	0.91	0.89	0.89	0.87	0.90	0.90	0.91	0.90	0.90
Accruals	0.89	0.91	0.86	0.90	0.83	0.92	0.87	0.89	0.88	0.84	0.86	0.94	0.88	0.87	0.91	0.96
Debt Issuance	0.86	0.91	0.85	0.91	0.86	0.93	0.92	0.90	0.92	0.89	0.91	0.90	0.93	0.93	0.89	0.92
Skewness	0.85	0.92	0.88	0.90	0.86	0.92	0.89	0.91	0.88	0.89	0.92	0.90	0.90	0.94	0.88	0.93
Seasonality	0.88	0.92	0.88	0.92	0.86	0.93	0.92	0.91	0.93	0.90	0.89	0.91	0.90	0.94	0.92	0.90

Note: I perform cojump tests between pairs of assets for each month in my sample. Using the resulting 300 test statistics per pair, I compute the fraction that are statistically significant using individual 5% level tests. The stock tickers correspond to Apple, Bank of America, Disney, General Electric, General Mills, International Business Machines, Johnson & Johnson, JP Morgan, Coca-Cola, 3M, Merck & Company, Microsoft, Procter & Gamble, Walmart, ExxonMobil, and Amazon.

Table A.3: FAMA-MACBETH REGRESSIONS – STANDARD RISK PREMIA

	<i>Specification</i>							
	CAPM		FF3		FF5		FF6	
FF MKT	5.18	(1.46)	5.40	(1.57)	5.70*	(1.67)	5.70*	(1.67)
FF SMB			0.17	(0.15)	0.42	(0.37)	0.71	(0.64)
FF HML			−0.28	(−0.17)	−0.59	(−0.37)	−0.49	(−0.31)
FF RMW					1.88*	(1.73)	1.81*	(1.68)
FF CMA					1.07	(1.12)	1.09	(1.15)
FF UMD							2.05	(0.91)
R^2	22.3%		25.3%		26.6%		27.3%	

Note: I report Fama-MacBeth estimates of the annualized risk premia (%) for each factor along with t-statistics in parentheses. The test assets include every portfolio in the factor zoo along with the yearly top 1000 stocks by market cap in each year. For each specification, I estimate monthly betas on daily data using a backwards-looking rolling window. Cross-sectional regressions are done on a monthly basis. The R^2 values report the time-series average of the R^2 estimates for each cross-sectional regression. The regressions include 3394 test assets and the risk premia are averaged over a time span of 24.9 years. The notation *, **, and *** refers to 90%, 95%, and 99% levels of significance respectively.

Table A.4: CTS-TIME FAMA-MACBETH REGRESSIONS – CONTINUOUS AND JUMP RISK PREMIA

		<i>Specification</i>							
		CAPM		FF3		FF5		FF6	
FF MKT	Continuous	2.27	(0.60)	0.44	(0.13)	1.72	(0.50)	1.55	(0.46)
	Jump	1.73	(0.68)	4.75**	(2.09)	5.38**	(2.50)	6.29***	(2.95)
FF SMB	Continuous			0.13	(0.07)	0.34	(0.18)	0.54	(0.29)
	Jump			0.30	(0.19)	1.33	(0.89)	1.77	(1.23)
FF HML	Continuous			−1.00	(−0.54)	−1.82	(−1.04)	−1.59	(−0.91)
	Jump			0.38	(0.29)	0.74	(0.60)	0.81	(0.68)
FF RMW	Continuous					4.36***	(3.67)	4.36***	(3.75)
	Jump					1.95**	(2.15)	1.76**	(1.97)
FF CMA	Continuous					0.31	(0.27)	0.84	(0.75)
	Jump					−0.32	(−0.38)	−0.09	(−0.10)
FF UMD	Continuous							1.55	(0.72)
	Jump							−2.11	(−1.29)
R^2		31.8%		36.1%		38.0%		39.1%	

Note: I report Cts-Time Fama-MacBeth estimates of the annualized risk premia (%) for each factor along with t-statistics in parentheses. The test assets include every portfolio in the factor zoo along with the top 1000 stocks by market cap in each year. The R^2 values report the time-series average of the R^2 estimates for each cross-sectional regression. The regressions include 3394 test assets and the risk premia are averaged over a time span of 24.9 years. The notation *, **, and *** refers to 90%, 95%, and 99% levels of significance respectively.

Table A.5: CLUSTER PORTFOLIOS – CONTINUOUS, SEMIJUMP, AND SEMIOVERNIGHT RISK PREMIA

<i>Risk Component Premia (Annualized %)</i>					
	Continuous	Positive Jump	Negative Jump	Positive Overnight	Negative Overnight
Profitability	3.39*** [0.95, 5.83]	-0.03 [-2.02, 1.96]	0.10 [-0.31, 0.51]	0.43 [-0.09, 0.94]	-0.30 [-2.49, 1.89]
Debt Issuance	0.53 [-0.96, 2.02]	1.22** [0.15, 2.29]	0.08 [-0.20, 0.37]	0.18 [-0.10, 0.46]	0.95* [-0.05, 1.95]
Momentum	2.63 [-1.38, 6.63]	2.62** [0.19, 5.05]	0.28 [-0.64, 1.20]	0.76 [-0.26, 1.77]	-1.55 [-3.52, 0.42]
Leverage	-1.92 [-5.04, 1.20]	0.46 [-1.55, 2.48]	-0.84** [-1.67, -0.00]	-0.47 [-1.16, 0.22]	0.04 [-1.49, 1.58]
Quality	2.81* [-0.04, 5.66]	0.37 [-1.67, 2.41]	0.02 [-0.60, 0.63]	0.23 [-0.56, 1.02]	-0.58 [-2.46, 1.29]
Low Risk	3.51 [-0.85, 7.88]	-0.41 [-2.48, 1.65]	0.34 [-0.64, 1.31]	1.16* [-0.05, 2.37]	-0.25 [-2.80, 2.29]
Investment	0.67 [-1.98, 3.31]	0.05 [-1.46, 1.57]	0.24 [-0.32, 0.81]	0.57* [-0.08, 1.22]	0.48 [-1.23, 2.20]
Profit Growth	-0.47 [-2.44, 1.51]	0.41 [-0.93, 1.75]	0.03 [-0.44, 0.50]	0.41* [-0.07, 0.89]	-0.33 [-1.38, 0.72]
Size	-0.09 [-3.78, 3.61]	1.75 [-0.64, 4.13]	-0.22 [-0.91, 0.46]	-0.02 [-0.62, 0.58]	-1.17 [-3.85, 1.51]
Value	-0.50 [-3.85, 2.84]	-0.25 [-2.32, 1.82]	-0.51 [-1.29, 0.28]	-0.32 [-1.02, 0.38]	-0.78 [-2.36, 0.81]
Accruals	0.15 [-1.89, 2.19]	0.08 [-1.36, 1.53]	-0.03 [-0.45, 0.38]	0.25 [-0.15, 0.66]	-0.96 [-2.45, 0.53]
Seasonality	1.49 [-2.30, 5.28]	-0.20 [-2.32, 1.92]	0.48 [-0.50, 1.47]	0.58 [-0.37, 1.54]	-0.93 [-3.20, 1.33]
Skewness	-0.96 [-3.52, 1.60]	0.34 [-1.30, 1.97]	-0.06 [-0.68, 0.55]	0.27 [-0.24, 0.78]	-0.59 [-2.19, 1.01]

Note: I report Cts-Time Fama-MacBeth estimates of the annualized risk premia (%) of each cluster's risk factors along with 95% confidence intervals in brackets. The test assets include every portfolio in the factor zoo, the cluster portfolios themselves, along with the top 1000 stocks by market cap in each year. For each specification, I estimate intraday continuous, intraday semijump, and up/down overnight betas for each spanning asset. All betas are estimated using a backwards-looking rolling window on 15-minute returns, with the jump betas using a yearly window and the overnight betas using a monthly window. Cross-sectional regressions are done on a monthly basis. The notation *, **, and *** refers to 90%, 95%, and 99% levels of significance respectively.

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